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Configuring positioning measurements and reports

Abstract

Apparatuses, methods, and systems are disclosed for configuring positioning measurements and reports. One apparatus in a mobile communication network includes a transceiver that receives, from a mobile wireless communication network, a positioning configuration defining a positioning configuration timeline and a measurement and processing time window for the UE The apparatus includes a processor that performs at least one positioning measurement for the UE according to the positioning processing timeline in response to receiving the positioning configuration. The transceiver sends a positioning measurement report comprising the at least one positioning measurement and measurement timeline performed of the at least one positioning measurement within the configured time window from the UE to the mobile wireless communication network.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims the benefit of U.S. Provisional Patent Application No. 63/076,683, entitled “UE PROCESSING ENHANCEMENTS FOR POSITIONING” and filed on Sep. 10, 2020, for Robin Thomas et al., and U.S. Provisional Patent Application No. 63/076,575, entitled “UE REPORTING ENHANCEMENTS FOR POSITIONING” and filed on Sep. 10, 2020, for Robin Thomas et al., which are incorporated herein by reference.

FIELD
(1) The subject matter disclosed herein relates generally to wireless communications and more

particularly relates to configuring positioning measurements and reports.

BACKGROUND

(2) In certain wireless communication systems, Radio Access Technology ("RAT") dependent positioning using 3GPP New Radio ("NR") technology is supported in specifications. The specifications may define certain requirements for positioning measurements and reports, which may include accuracy, latency, and reliability positioning requirements.

BRIEF SUMMARY

(3) Disclosed are procedures for configuring positioning measurements and reports. The procedures may be implemented by apparatus, systems, methods, or computer program products.

(4) In one embodiment, an apparatus includes a transceiver that receives, from a mobile wireless communication network, a positioning configuration defining a positioning configuration timeline and a measurement and processing time window for the UE. The positioning configuration may include a timeline duration defining when to start performing measurements, a set of positioning measurements to be taken within the configured time window, and a window duration for measuring and processing requested position-relation measurements for the UE according to the positioning processing timeline.

(5) In one embodiment, the apparatus includes a processor that performs at least one positioning measurement for the UE according to the positioning processing timeline in response to receiving the positioning configuration. In certain embodiments, the transceiver sends a positioning measurement report comprising the at least one positioning measurement and measurement timeline performed of the at least one positioning measurement within the configured time window from the UE to the mobile wireless communication network.

(6) In one embodiment, another apparatus includes a transceiver that sends, to a User Equipment ("UE") device, a positioning configuration defining a positioning configuration timeline and a measurement and processing time window for the UE. In one embodiment, the positioning configuration comprising a timeline duration defining when to start performing measurements, a set of positioning measurements to be taken within the configured time window, and a window duration for measuring and processing requested position-relation measurements for the UE according to the positioning processing timeline. In one embodiment, the transceiver receives, from the UE device, a positioning measurement report comprising the at least one positioning measurement and measurement timeline of the at least one positioning measurement performed within the configured time window.

(7) In one embodiment, another apparatus includes a transceiver that receives, from a mobile wireless communication network, an uplink ("UL") configured grant configuration based on criteria associated with at least one of a positioning latency budget and a positioning processing timeline for the UE. In some embodiments, the apparatus includes a processor that performs at least one positioning measurement for the UE according to the positioning processing timeline and generates a positioning measurement report comprising the at least one positioning measurement.

(8) In certain embodiments, the transceiver sends the positioning measurement report to the mobile wireless communication network using the UL configured grant configuration based on an availability of positioning-related reference signal measurements within at least one of the positioning latency budget and the positioning processing timeline.

(9) In one embodiment, another embodiment includes a transceiver that sends, to a User Equipment ("UE") device, an uplink ("UL") configured grant configuration based on criteria associated with at least one of a positioning latency budget and a positioning processing timeline for the UE and receives a positioning measurement report from the UE device using the UL configured grant configuration based on an availability of positioning-related reference signal measurements within at least one of the positioning latency budget and the positioning processing timeline.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) A more particular description of the embodiments briefly described above will be rendered by reference to specific embodiments that are illustrated in the appended drawings. Understanding that these drawings depict only some embodiments and are not therefore to be considered to be limiting of scope, the embodiments will be described and explained with additional specificity and detail through the use of the accompanying drawings, in which:

(2) FIG. **1** is a schematic block diagram illustrating one embodiment of a wireless communication system for configuring positioning measurements and reports;

(3) FIG. **2** is a block diagram illustrating one embodiment of a 5G New Radio ("NR") protocol stack;

(4) FIG. **3** is a diagram illustrating one embodiment of NR Beam-based Positioning;

(5) FIG. **4** is a diagram illustrating one embodiment of DL-TDOA Assistance Data;

(6) FIG. **5** is a diagram illustrating one embodiment of DL-TDOA Measurement Report;

(7) FIG. **6** is a diagram illustrating one embodiment of UE-assisted Positioning for configuring, measuring, and processing positioning measurements and sending reports;

(8) FIG. **7** is a diagram illustrating one embodiment of UE-based Positioning for configuring, measuring, and processing positioning measurements and sending reports;

(9) FIG. **8** is a diagram illustrating one embodiment of UE positioning processing timeline with MG configuration for configuring, measuring, and processing positioning measurements and sending reports;

(10) FIG. **9** is a diagram illustrating one embodiment of dynamic positioning measurement reporting based on UL CG for configuring, measuring, and processing positioning measurements and sending reports;

(11) FIG. **10** is a block diagram illustrating one embodiment of a user equipment apparatus that may be used for configuring, measuring, and processing positioning measurements and sending reports;

(12) FIG. **11** is a block diagram illustrating one embodiment of a network equipment apparatus that may be used for configuring, measuring, and processing positioning measurements and sending reports;

(13) FIG. **12** is a block diagram illustrating one embodiment of a first method for configuring, measuring, and processing positioning measurements and sending reports;

(14) FIG. **13** is a block diagram illustrating one embodiment of a second method for configuring, measuring, and processing positioning measurements and sending reports;

(15) FIG. **14** is a block diagram illustrating one embodiment of a third method for configuring, measuring, and processing positioning measurements and sending reports;

(16) FIG. **15** is a block diagram illustrating one embodiment of a fourth method for configuring, measuring, and processing positioning measurements and sending reports.

DETAILED DESCRIPTION

(17) As will be appreciated by one skilled in the art, aspects of the embodiments may be embodied as a system, apparatus, method, or program product. Accordingly, embodiments may take the form of an entirely hardware embodiment, an entirely software embodiment (including firmware, resident software, micro-code, etc.) or an embodiment combining software and hardware aspects.

(18) For example, the disclosed embodiments may be implemented as a hardware circuit comprising custom very-large-scale integration ("VLSI") circuits or gate arrays, off-the-shelf semiconductors such as logic chips, transistors, or other discrete components. The disclosed embodiments may also be implemented in programmable hardware devices such as field programmable gate arrays, programmable array logic, programmable logic devices, or the like. As

another example, the disclosed embodiments may include one or more physical or logical blocks of executable code which may, for instance, be organized as an object, procedure, or function.

(19) Furthermore, embodiments may take the form of a program product embodied in one or more computer readable storage devices storing machine readable code, computer readable code, and/or program code, referred hereafter as code. The storage devices may be tangible, non-transitory, and/or non-transmission. The storage devices may not embody signals. In a certain embodiment, the storage devices only employ signals for accessing code.

(20) Any combination of one or more computer readable medium may be utilized. The computer readable medium may be a computer readable storage medium. The computer readable storage medium may be a storage device storing the code. The storage device may be, for example, but not limited to, an electronic, magnetic, optical, electromagnetic, infrared, holographic, micromechanical, or semiconductor system, apparatus, or device, or any suitable combination of the foregoing.

(21) More specific examples (a non-exhaustive list) of the storage device would include the following: an electrical connection having one or more wires, a portable computer diskette, a hard disk, a random-access memory (“RAM”), a read-only memory (“ROM”), an erasable programmable read-only memory (“EPROM” or Flash memory), a portable compact disc read-only memory (“CD-ROM”), an optical storage device, a magnetic storage device, or any suitable combination of the foregoing. In the context of this document, a computer readable storage medium may be any tangible medium that can contain or store a program for use by or in connection with an instruction execution system, apparatus, or device.

(22) Code for carrying out operations for embodiments may be any number of lines and may be written in any combination of one or more programming languages including an object-oriented programming language such as Python, Ruby, Java, Smalltalk, C++, or the like, and conventional procedural programming languages, such as the “C” programming language, or the like, and/or machine languages such as assembly languages. The code may execute entirely on the user's computer, partly on the user's computer, as a stand-alone software package, partly on the user's computer and partly on a remote computer or entirely on the remote computer or server. In the latter scenario, the remote computer may be connected to the user's computer through any type of network, including a local area network (“LAN”), wireless LAN (“WLAN”), or a wide area network (“WAN”), or the connection may be made to an external computer (for example, through the Internet using an Internet Service Provider (“ISP”)).

(23) Furthermore, the described features, structures, or characteristics of the embodiments may be combined in any suitable manner. In the following description, numerous specific details are provided, such as examples of programming, software modules, user selections, network transactions, database queries, database structures, hardware modules, hardware circuits, hardware chips, etc., to provide a thorough understanding of embodiments. One skilled in the relevant art will recognize, however, that embodiments may be practiced without one or more of the specific details, or with other methods, components, materials, and so forth. In other instances, well-known structures, materials, or operations are not shown or described in detail to avoid obscuring aspects of an embodiment.

(24) Reference throughout this specification to “one embodiment,” “an embodiment,” or similar language means that a particular feature, structure, or characteristic described in connection with the embodiment is included in at least one embodiment. Thus, appearances of the phrases “in one embodiment,” “in an embodiment,” and similar language throughout this specification may, but do not necessarily, all refer to the same embodiment, but mean “one or more but not all embodiments” unless expressly specified otherwise. The terms “including,” “comprising,” “having,” and variations thereof mean “including but not limited to,” unless expressly specified otherwise. An enumerated listing of items does not imply that any or all of the items are mutually exclusive, unless expressly specified otherwise. The terms “a,” “an,” and “the” also refer to “one or more”

unless expressly specified otherwise.

(25) As used herein, a list with a conjunction of “and/or” includes any single item in the list or a combination of items in the list. For example, a list of A, B and/or C includes only A, only B, only C, a combination of A and B, a combination of B and C, a combination of A and C or a combination of A, B and C. As used herein, a list using the terminology “one or more of” includes any single item in the list or a combination of items in the list. For example, one or more of A, B and C includes only A, only B, only C, a combination of A and B, a combination of B and C, a combination of A and C or a combination of A, B and C. As used herein, a list using the terminology “one of” includes one and only one of any single item in the list. For example, “one of A, B and C” includes only A, only B or only C and excludes combinations of A, B and C. As used herein, “a member selected from the group consisting of A, B, and C,” includes one and only one of A, B, or C, and excludes combinations of A, B, and C. As used herein, “a member selected from the group consisting of A, B, and C and combinations thereof” includes only A, only B, only C, a combination of A and B, a combination of B and C, a combination of A and C or a combination of A, B and C.

(26) Aspects of the embodiments are described below with reference to schematic flowchart diagrams and/or schematic block diagrams of methods, apparatuses, systems, and program products according to embodiments. It will be understood that each block of the schematic flowchart diagrams and/or schematic block diagrams, and combinations of blocks in the schematic flowchart diagrams and/or schematic block diagrams, can be implemented by code. This code may be provided to a processor of a general-purpose computer, special purpose computer, or other programmable data processing apparatus to produce a machine, such that the instructions, which execute via the processor of the computer or other programmable data processing apparatus, create means for implementing the functions/acts specified in the flowchart diagrams and/or block diagrams.

(27) The code may also be stored in a storage device that can direct a computer, other programmable data processing apparatus, or other devices to function in a particular manner, such that the instructions stored in the storage device produce an article of manufacture including instructions which implement the function/act specified in the flowchart diagrams and/or block diagrams.

(28) The code may also be loaded onto a computer, other programmable data processing apparatus, or other devices to cause a series of operational steps to be performed on the computer, other programmable apparatus, or other devices to produce a computer implemented process such that the code which execute on the computer or other programmable apparatus provide processes for implementing the functions/acts specified in the flowchart diagrams and/or block diagrams.

(29) The flowchart diagrams and/or block diagrams in the Figures illustrate the architecture, functionality, and operation of possible implementations of apparatuses, systems, methods, and program products according to various embodiments. In this regard, each block in the flowchart diagrams and/or block diagrams may represent a module, segment, or portion of code, which includes one or more executable instructions of the code for implementing the specified logical function(s).

(30) It should also be noted that, in some alternative implementations, the functions noted in the block may occur out of the order noted in the Figures. For example, two blocks shown in succession may, in fact, be executed substantially concurrently, or the blocks may sometimes be executed in the reverse order, depending upon the functionality involved. Other steps and methods may be conceived that are equivalent in function, logic, or effect to one or more blocks, or portions thereof, of the illustrated Figures.

(31) Although various arrow types and line types may be employed in the flowchart and/or block diagrams, they are understood not to limit the scope of the corresponding embodiments. Indeed, some arrows or other connectors may be used to indicate only the logical flow of the depicted

embodiment. For instance, an arrow may indicate a waiting or monitoring period of unspecified duration between enumerated steps of the depicted embodiment. It will also be noted that each block of the block diagrams and/or flowchart diagrams, and combinations of blocks in the block diagrams and/or flowchart diagrams, can be implemented by special purpose hardware-based systems that perform the specified functions or acts, or combinations of special purpose hardware and code.

(32) The description of elements in each figure may refer to elements of proceeding figures. Like numbers refer to like elements in all figures, including alternate embodiments of like elements.

(33) Generally, the present disclosure describes systems, methods, and apparatuses for configuring positioning measurements and reports. In certain embodiments, the methods may be performed using computer code embedded on a computer-readable medium. In certain embodiments, an apparatus or system may include a computer-readable medium containing computer-readable code which, when executed by a processor, causes the apparatus or system to perform at least a portion of the below described solutions.

(34) In one embodiment, the subject matter disclosed herein describes enhancements that address issues related to the UE processing timeline of downlink (“DL”)-positioning reference signals (“PRS”) and/or other related positioning-related reference signals. Different processing functionalities can address different latency and location accuracy requirements. In certain embodiments, there have been no proposed enhancements regarding UE processing timeline configurations for RAT-dependent positioning procedures including measurement and reporting of positioning-related reference signals.

(35) In one embodiment, to satisfy the requirements of low latency positioning, it would be beneficial to define different UE processing timelines depending on target-UE's capabilities. A target-UE(s) in the context of this disclosure may refer to the UE(s) to be localized. The present disclosure aims to tackle this UE processing timeline issue for positioning and introduces new functionality to enable low-latency positioning. Furthermore, the UE positioning processing timeline may be applicable to UE-assisted and UE-based positioning methods. The subject matter herein also describes management of the UE processing timeline when a measurement gap is configured to the target-UE for the positioning purposes, which can also have an impact on the UE processing load.

(36) In further embodiments, the subject matter disclosed herein describes enhancements that address the issues related to the high measurement and reporting latency of DL-PRS and/or other related positioning-related reference signals. Dynamic Layer-½ signaling can reduce the time required to process a measurement and report it to the location server. In certain embodiments, higher-layer non-access stratum (“NAS”) LPP signaling is used to configure RAT-dependent measurement and reporting of positioning-related reference signals such as DL-PRS, which can be inefficient and incur high latency.

(37) The delay between a UE receiving a DL-PRS measurement reporting configuration (e.g., in the case of UE-assisted positioning) or location estimate request (e.g., in the case of UE-based positioning) and the UE providing the said measurement report/location estimate to the location server, e.g., LMF, can be further optimized in order to reduce the overall positioning latency (Time-To-First-Fix).

(38) The present disclosure, in one embodiment, provides mechanisms to enable dynamic signaling to reduce the overall positioning latency when processing the measurements and transmitting corresponding reports to location server. Uplink (“UL”) configured grants for positioning may reduce the report transmission time, depending on the availability. Priority indications for measurement processing may also assist in ranking which measurement reports can be prioritized based on the UL resource availability. In order to enable efficient and low latency reporting of positioning-related reference signals, certain measurements may also be dropped based on a set of criteria, which is detailed in the disclosure.

(39) For Release 17 ("Rel-17") of the 3GPP specification, the different positioning requirements are especially stringent with respect to accuracy, latency, and reliability. Table 1 shows positioning performance requirements for different scenarios in an Industrial IoT ("IIoT") or indoor factory setting. Note that augmented reality in smart factories may have a heading positioning performance requirements of <0.17 radians and mobile control panels with safety functions in smart factories (within factory danger zones) may have a heading positioning performance requirements of <0.54 radians.
(40) TABLE-US-00001 TABLE 1 IIoT Positioning Performance Requirements Latency for position Corresponding Horizontal Vertical estimation UE Positioning Scenario accuracy accuracy Availability of UE Speed Service Level Mobile control panels with <5 m <3 m 90% <5 s N/A Service Level 2 safety functions (non- danger zones) Process automation - plant <1 m <3 m 90% <2 s <30 km/h Service Level 3 asset management Flexible, modular <1 m N/A 99% 1 s <30 km/h Service Level 3 assembly area in smart (relative factories (for tracking of positioning) tools at the work-place location) Augmented reality in smart <1 m <3 m 99% <15 ms <10 km/h Service Level 4 factories Mobile control panels with <1 m <3 m 99.9%.sup.   <1 s N/A Service Level 4 safety functions in smart factories (within factory danger zones) Flexible, modular <50 cm <3 m 99% 1 s <30 km/h Service Level 5 assembly area in smart factories (for autonomous vehicles, only for monitoring proposes) Inbound logistics for <30 cm (if <3 m 99.9%.sup.   10 ms <30 km/h Service Level 6 manufacturing (for driving supported trajectories (if supported by further by further sensors like sensors like camera, GNSS, IMU) of camera, indoor autonomous driving GNSS, systems)) IMU) Inbound logistics for <20 cm <20 cm 99% <1 s <30 km/h Service Level 7 manufacturing (for storage of goods)
(41) The present disclosure provides enhancements to reduce the UE processing timeline for positioning-related reference signals with an emphasis on low-latency positioning. Note that for the purposes of this disclosure, a positioning-related reference signal may refer to a reference signal used for positioning procedures and/or purposes to estimate a target-UE's location, e.g., PRS or based on existing reference signals such as sounding reference signal ("SRS"). In one embodiment, a target-UE can be referred to as the device/entity to be localized.
(42) In one embodiment, a method is disclosed to define processing timeline(s) for positioning-related reference signals based on at least one of a combination of following criteria: i. One or more UE capabilities such as for an enhanced mobile broadband ("eMBB") device or an ultra-reliable and low-latency communications ("URLLC") device including: 1. Latency 2. Device efficiency ii. Positioning accuracy requirements iii. Number of positioning measurement related quantities to be reported iv. Type of measurement to be processed
(43) In one embodiment, the present disclosure establishes and specifies requirements for processing position related measurements from a UE capability point of view where the different positioning timeline configurations accommodate different positioning latency requirements and UE capabilities.
(44) In one embodiment, a method is disclosed to determine the appropriate resources for using the configured grant or multiple UL grants for reporting the PRS-based measurements, e.g., UL resources required to report the ProvideLocation message. In such an embodiment, the instance from the time the measurements are available to report to obtaining the UL resource can be adapted and configured to enable low latency positioning.
(45) In certain embodiments, a method is disclosed for prioritization of PRS measurement reports based on the availability of UL resources, UE processing timeline, and accuracy requirements. In such an embodiment, priority handling mechanisms enable the LMF to acquire the positioning measurements within the required duration based on configured positioning method, especially in the case of UE-assisted positioning methods.
(46) In further embodiments, a method is disclosed for dropping measurement reports, based on a set of criteria, e.g., reports that are not transmitted within the required timeline and positioning

latency budget. In such an embodiment, obsolete measurements do not need to be stored in the buffer, thus increasing efficiency of handling UE measurements.

(47) FIG. **1** depicts a wireless communication system **100** for configuring positioning measurements and reports, according to embodiments of the disclosure. In one embodiment, the wireless communication system **100** includes at least one remote unit **105**, a radio access network (“RAN”) **120**, and a mobile core network **140**. The RAN **120** and the mobile core network **140** form a mobile communication network. The RAN **120** may be composed of a base unit **121** with which the remote unit **105** communicates using wireless communication links **123**. Even though a specific number of remote units **105**, base units **121**, wireless communication links **123**, RANs **120**, and mobile core networks **140** are depicted in FIG. **1**, one of skill in the art will recognize that any number of remote units **105**, base units **121**, wireless communication links **123**, RANs **120**, and mobile core networks **140** may be included in the wireless communication system **100**.

(48) In one implementation, the RAN **120** is compliant with the 5G system specified in the Third Generation Partnership Project (“3GPP”) specifications. For example, the RAN **120** may be a Next Generation Radio Access Network (“NG-RAN”), implementing New Radio (“NR”) Radio Access Technology (“RAT”) and/or Long-Term Evolution (“LTE”) RAT. In another example, the RAN **120** may include non-3GPP RAT (e.g., Wi-Fi® or Institute of Electrical and Electronics Engineers (“IEEE”) 802.11-family compliant WLAN). In another implementation, the RAN **120** is compliant with the LTE system specified in the 3GPP specifications. More generally, however, the wireless communication system **100** may implement some other open or proprietary communication network, for example Worldwide Interoperability for Microwave Access (“WiMAX”) or IEEE 802.16-family standards, among other networks. The present disclosure is not intended to be limited to the implementation of any particular wireless communication system architecture or protocol.

(49) In one embodiment, the remote units **105** may include computing devices, such as desktop computers, laptop computers, personal digital assistants (“PDAs”), tablet computers, smart phones, smart televisions (e.g., televisions connected to the Internet), smart appliances (e.g., appliances connected to the Internet), set-top boxes, game consoles, security systems (including security cameras), vehicle on-board computers, network devices (e.g., routers, switches, modems), or the like. In some embodiments, the remote units **105** include wearable devices, such as smart watches, fitness bands, optical head-mounted displays, or the like. Moreover, the remote units **105** may be referred to as the UEs, subscriber units, mobiles, mobile stations, users, terminals, mobile terminals, fixed terminals, subscriber stations, user terminals, wireless transmit/receive unit (“WTRU”), a device, or by other terminology used in the art. In various embodiments, the remote unit **105** includes a subscriber identity and/or identification module (“SIM”) and the mobile equipment (“ME”) providing mobile termination functions (e.g., radio transmission, handover, speech encoding and decoding, error detection and correction, signaling and access to the SIM). In certain embodiments, the remote unit **105** may include a terminal equipment (“TE”) and/or be embedded in an appliance or device (e.g., a computing device, as described above).

(50) The remote units **105** may communicate directly with one or more of the base units **121** in the RAN **120** via uplink (“UL”) and downlink (“DL”) communication signals. Furthermore, the UL and DL communication signals may be carried over the wireless communication links **123**. Here, the RAN **120** is an intermediate network that provides the remote units **105** with access to the mobile core network **140**. As described in greater detail below, the base unit(s) **121** may provide a cell operating using a first frequency range and/or a cell operating using a second frequency range.

(51) In some embodiments, the remote units **105** communicate with an application server **151** via a network connection with the mobile core network **140**. For example, an application **107** (e.g., web browser, media client, telephone and/or Voice-over-Internet-Protocol (“VoIP”) application) in a remote unit **105** may trigger the remote unit **105** to establish a protocol data unit (“PDU”) session (or other data connection) with the mobile core network **140** via the RAN **120**. The mobile core

network **140** then relays traffic between the remote unit **105** and the application server **151** in the packet data network **150** using the PDU session. The PDU session represents a logical connection between the remote unit **105** and the User Plane Function ("UPF") **141**.

(52) In order to establish the PDU session (or PDN connection), the remote unit **105** must be registered with the mobile core network **140** (also referred to as "attached to the mobile core network" in the context of a Fourth Generation ("4G") system). Note that the remote unit **105** may establish one or more PDU sessions (or other data connections) with the mobile core network **140**. As such, the remote unit **105** may have at least one PDU session for communicating with the packet data network **150**. The remote unit **105** may establish additional PDU sessions for communicating with other data networks and/or other communication peers.

(53) In the context of a 5G system ("5GS"), the term "PDU Session" refers to a data connection that provides end-to-end ("E2E") user plane ("UP") connectivity between the remote unit **105** and a specific Data Network ("DN") through the UPF **141**. A PDU Session supports one or more Quality of Service ("QoS") Flows. In certain embodiments, there may be a one-to-one mapping between a QoS Flow and a QoS profile, such that all packets belonging to a specific QoS Flow have the same 5G QoS Identifier ("5QI").

(54) In the context of a 4G/LTE system, such as the Evolved Packet System ("EPS"), a Packet Data Network ("PDN") connection (also referred to as EPS session) provides E2E UP connectivity between the remote unit and a PDN. The PDN connectivity procedure establishes an EPS Bearer, i.e., a tunnel between the remote unit **105** and a Packet Gateway ("PGW", not shown) in the mobile core network **140**. In certain embodiments, there is a one-to-one mapping between an EPS Bearer and a QoS profile, such that all packets belonging to a specific EPS Bearer have the same QoS Class Identifier ("QCI").

(55) The base units **121** may be distributed over a geographic region. In certain embodiments, a base unit **121** may also be referred to as an access terminal, an access point, a base, a base station, a Node-B ("NB"), an Evolved Node B (abbreviated as eNodeB or "eNB," also known as Evolved Universal Terrestrial Radio Access Network ("E-UTRAN") Node B), a 5G/NR Node B ("gNB"), a Home Node-B, a relay node, a RAN node, or by any other terminology used in the art. The base units **121** are generally part of a RAN, such as the RAN **120**, that may include one or more controllers communicably coupled to one or more corresponding base units **121**. These and other elements of radio access network are not illustrated but are well known generally by those having ordinary skill in the art. The base units **121** connect to the mobile core network **140** via the RAN **120**.

(56) The base units **121** may serve a number of remote units **105** within a serving area, for example, a cell or a cell sector, via a wireless communication link **123**. The base units **121** may communicate directly with one or more of the remote units **105** via communication signals. Generally, the base units **121** transmit DL communication signals to serve the remote units **105** in the time, frequency, and/or spatial domain. Furthermore, the DL communication signals may be carried over the wireless communication links **123**. The wireless communication links **123** may be any suitable carrier in licensed or unlicensed radio spectrum. The wireless communication links **123** facilitate communication between one or more of the remote units **105** and/or one or more of the base units **121**. Note that during NR operation on unlicensed spectrum (referred to as "NR-U"), the base unit **121** and the remote unit **105** communicate over unlicensed (i.e., shared) radio spectrum.

(57) In one embodiment, the mobile core network **140** is a 5GC or an Evolved Packet Core ("EPC"), which may be coupled to a packet data network **150**, like the Internet and private data networks, among other data networks. A remote unit **105** may have a subscription or other account with the mobile core network **140**. In various embodiments, each mobile core network **140** belongs to a single mobile network operator ("MNO"). The present disclosure is not intended to be limited to the implementation of any particular wireless communication system architecture or protocol.

(58) The mobile core network **140** includes several network functions ("NFs"). As depicted, the mobile core network **140** includes at least one UPF **141**. The mobile core network **140** also includes multiple control plane ("CP") functions including, but not limited to, an Access and Mobility Management Function ("AMF") **143** that serves the RAN **120**, a Session Management Function ("SMF") **145**, a Location Management Function ("LMF") **144**, a Unified Data Management function ("UDM ") and a User Data Repository ("UDR"). Although specific numbers and types of network functions are depicted in FIG. **1**, one of skill in the art will recognize that any number and type of network functions may be included in the mobile core network **140**.
(59) The UPF(s) **141** is/are responsible for packet routing and forwarding, packet inspection, QoS handling, and external PDU session for interconnecting Data Network (DN), in the 5G architecture. The AMF **143** is responsible for termination of NAS signaling, NAS ciphering & integrity protection, registration management, connection management, mobility management, access authentication and authorization, security context management. The SMF **145** is responsible for session management (i.e., session establishment, modification, release), remote unit (i.e., UE) IP address allocation & management, DL data notification, and traffic steering configuration of the UPF **141** for proper traffic routing.
(60) The LMF **144** receives positioning measurements or estimates from RAN **120** and the remote unit **105** (e.g., via the AMF **143**) and computes the position of the remote unit **105**. The UDM is responsible for generation of Authentication and Key Agreement ("AKA") credentials, user identification handling, access authorization, subscription management. The UDR is a repository of subscriber information and may be used to service a number of network functions. For example, the UDR may store subscription data, policy-related data, subscriber-related data that is permitted to be exposed to third party applications, and the like. In some embodiments, the UDM is co-located with the UDR, depicted as combined entity "UDM/UDR" **149**.
(61) In various embodiments, the mobile core network **140** may also include a Policy Control Function ("PCF") (which provides policy rules to CP functions), a Network Repository Function ("NRF") (which provides Network Function ("NF") service registration and discovery, enabling NFs to identify appropriate services in one another and communicate with each other over Application Programming Interfaces ("APIs")), a Network Exposure Function ("NEF") (which is responsible for making network data and resources easily accessible to customers and network partners), an Authentication Server Function ("AUSF"), or other NFs defined for the 5GC. When present, the AUSF may act as an authentication server and/or authentication proxy, thereby allowing the AMF **143** to authenticate a remote unit **105**. In certain embodiments, the mobile core network **140** may include an authentication, authorization, and accounting ("AAA") server.
(62) In various embodiments, the mobile core network **140** supports different types of mobile data connections and different types of network slices, wherein each mobile data connection utilizes a specific network slice. Here, a "network slice" refers to a portion of the mobile core network **140** optimized for a certain traffic type or communication service. For example, one or more network slices may be optimized for enhanced mobile broadband ("eMBB") service. As another example, one or more network slices may be optimized for ultra-reliable low-latency communication ("URLLC") service. In other examples, a network slice may be optimized for machine-type communication ("MTC") service, massive MTC ("mMTC") service, Internet-of-Things ("IoT") service. In yet other examples, a network slice may be deployed for a specific application service, a vertical service, a specific use case, etc.
(63) A network slice instance may be identified by a single-network slice selection assistance information ("S-NSSAI") while a set of network slices for which the remote unit **105** is authorized to use is identified by network slice selection assistance information ("NSSAI"). Here, "NSSAI" refers to a vector value including one or more S-NSSAI values. In certain embodiments, the various network slices may include separate instances of network functions, such as the SMF **145** and UPF **141**. In some embodiments, the different network slices may share some common

network functions, such as the AMF **143**. The different network slices are not shown in FIG. **1** for ease of illustration, but their support is assumed.

(64) As discussed in greater detail below, the remote unit **105** receives a positioning measurement configuration **125** from the network (e.g., from the LMF **144** via RAN **120**), including a positioning processing timeline for the remote unit **105** based on the remote unit's capabilities. The remote unit **105** performs positioning measurements, as described in greater detail below, and sends a positioning report to the LMF **144**.

(65) While FIG. **1** depicts components of a 5G RAN and a 5G core network, the described embodiments for configuring positioning measurements and reports apply to other types of communication networks and RATs, including IEEE 802.11 variants, Global System for Mobile Communications ("GSM", i.e., a 2G digital cellular network), General Packet Radio Service ("GPRS"), Universal Mobile Telecommunications System ("UMTS"), LTE variants, CDMA 2000, Bluetooth, ZigBee, Sigfox, and the like.

(66) Moreover, in an LTE variant where the mobile core network **140** is an EPC, the depicted network functions may be replaced with appropriate EPC entities, such as a Mobility Management Entity ("MME"), a Serving Gateway ("SGW"), a PGW, a Home Subscriber Server ("HSS"), and the like. For example, the AMF **143** may be mapped to an MME, the SMF **145** may be mapped to a control plane portion of a PGW and/or to an MME, the UPF **141** may be mapped to an SGW and a user plane portion of the PGW, the UDM/UDR **149** may be mapped to an HSS, etc.

(67) In the following descriptions, the term "RAN node" is used for the base station but it is replaceable by any other radio access node, e.g., gNB, ng-eNB, eNB, Base Station ("BS"), Access Point ("AP"), etc. Further, the operations are described mainly in the context of 5G NR. However, the proposed solutions/methods are also equally applicable to other mobile communication systems supporting configuring positioning measurements and reports.

(68) FIG. **2** depicts a NR protocol stack **200**, according to embodiments of the disclosure. While FIG. **2** shows the UE **205**, the RAN node **210** and an AMF **215** in a 5G core network ("5GC"), these are representative of a set of remote units **105** interacting with a base unit **121** and a mobile core network **140**. As depicted, the protocol stack **200** comprises a User Plane protocol stack **201** and a Control Plane protocol stack **203**. The User Plane protocol stack **201** includes a physical ("PHY") layer **220**, a Medium Access Control ("MAC") sublayer **225**, the Radio Link Control ("RLC") sublayer **230**, a Packet Data Convergence Protocol ("PDCP") sublayer **235**, and Service Data Adaptation Protocol ("SDAP") layer **240**. The Control Plane protocol stack **203** includes a physical layer **220**, a MAC sublayer **225**, a RLC sublayer **230**, and a PDCP sublayer **235**. The Control Plane protocol stack **203** also includes a Radio Resource Control ("RRC") layer **245** and a Non-Access Stratum ("NAS") layer **250**.

(69) The AS layer (also referred to as "AS protocol stack") for the User Plane protocol stack **201** consists of at least SDAP, PDCP, RLC and MAC sublayers, and the physical layer. The AS layer for the Control Plane protocol stack **203** consists of at least RRC, PDCP, RLC and MAC sublayers, and the physical layer. The Layer-2 ("L2") is split into the SDAP, PDCP, RLC and MAC sublayers. The Layer-3 ("L3") includes the RRC sublayer **245** and the NAS layer **250** for the control plane and includes, e.g., an Internet Protocol ("IP") layer and/or PDU Layer (not depicted) for the user plane. L1 and L2 are referred to as "lower layers," while L3 and above (e.g., transport layer, application layer) are referred to as "higher layers" or "upper layers."

(70) The physical layer **220** offers transport channels to the MAC sublayer **225**. The physical layer **220** may perform a Clear Channel Assessment and/or Listen-Before-Talk ("CCA/LBT") procedure using energy detection thresholds, as described herein. In certain embodiments, the physical layer **220** may send a notification of UL Listen-Before-Talk ("LBT") failure to a MAC entity at the MAC sublayer **225**. The MAC sublayer **225** offers logical channels to the RLC sublayer **230**. The RLC sublayer **230** offers RLC channels to the PDCP sublayer **235**. The PDCP sublayer **235** offers radio bearers to the SDAP sublayer **240** and/or RRC layer **245**. The SDAP sublayer **240** offers QoS

flows to the core network (e.g., 5GC). The RRC layer **245** provides for the addition, modification, and release of Carrier Aggregation and/or Dual Connectivity. The RRC layer **245** also manages the establishment, configuration, maintenance, and release of Signaling Radio Bearers ("SRBs") and Data Radio Bearers ("DRBs").

(71) The NAS layer **250** is between the UE **205** and the 5GC **215**. NAS messages are passed transparently through the RAN. The NAS layer **250** is used to manage the establishment of communication sessions and for maintaining continuous communications with the UE **205** as it moves between different cells of the RAN. In contrast, the AS layer is between the UE **205** and the RAN (i.e., RAN node **210**) and carries information over the wireless portion of the network.

(72) In one embodiment, the following RAT-dependent positioning techniques may be supported by the system **100**:

(73) DL-TDoA: The DL TDOA positioning method makes use of the DL RS Time Difference ("RSTD") (and optionally DL PRS RS Received Power ("RSRP") of DL PRS RS Received Quality ("RSRQ")) of downlink signals received from multiple TPs, at the UE **205** (i.e., remote unit **105**). The UE **205** measures the DL RSTD (and optionally DL PRS RSRP) of the received signals using assistance data received from the positioning server, and the resulting measurements are used along with other configuration information to locate the UE **205** in relation to the neighboring Transmission Points ("TPs").

(74) DL-AoD: The DL Angle of Departure ("AoD") positioning method makes use of the measured DL PRS RSRP of downlink signals received from multiple TPs, at the UE **205**. The UE **205** measures the DL PRS RSRP of the received signals using assistance data received from the positioning server, and the resulting measurements are used along with other configuration information to locate the UE **205** in relation to the neighboring TPs.

(75) Multi-RTT: The Multiple-Round Trip Time ("Multi-RTT") positioning method makes use of the UE Receive-Transmit ("Rx-Tx") measurements and DL PRS RSRP of downlink signals received from multiple TRPs, measured by the UE **205** and the gNB Rx-Tx measurements (i.e., measured by RAN node **210**) and UL SRS-RSRP at multiple TRPs of uplink signals transmitted from UE **205**.

(76) The UE **205** measures the UE Rx-Tx measurements (and optionally DL PRS RSRP of the received signals) using assistance data received from the positioning server, and the TRPs measure the gNB Rx-Tx measurements (and optionally UL SRS-RSRP of the received signals) using assistance data received from the positioning server. The measurements are used to determine the Round Trip Time ("RTT") at the positioning server which are used to estimate the location of the UE **205**.

(77) E-CID/NR E-CID: Enhanced Cell ID (CID) positioning method, the position of a UE **205** is estimated with the knowledge of its serving ng-eNB, gNB and cell and is based on LTE signals. The information about the serving ng-eNB, gNB and cell may be obtained by paging, registration, or other methods. NR Enhanced Cell ID (NR E CID) positioning refers to techniques which use additional UE measurements and/or NR radio resource and other measurements to improve the UE location estimate using NR signals.

(78) Although NR E-CID positioning may utilize some of the same measurements as the measurement control system in the RRC protocol, the UE **205** generally is not expected to make additional measurements for the sole purpose of positioning; i.e., the positioning procedures do not supply a measurement configuration or measurement control message, and the UE **205** reports the measurements that it has available rather than being required to take additional measurement actions.

(79) UL-TDoA: The UL TDOA positioning method makes use of the UL TDOA (and optionally UL SRS-RSRP) at multiple RPs of uplink signals transmitted from the UE **205**. The RPs measure the UL TDOA (and optionally UL SRS-RSRP) of the received signals using assistance data received from the positioning server, and the resulting measurements are used along with other

configuration information to estimate the location of the UE **205**.
(80) UL-AoA: The UL Angle of Arrival ("AoA") positioning method makes use of the measured azimuth and the zenith angles of arrival at multiple RPs of uplink signals transmitted from the UE **205**. The RPs measure A-AoA and Z-AoA of the received signals using assistance data received from the positioning server, and the resulting measurements are used along with other configuration information to estimate the location of the UE **205**.
(81) Some UE positioning method supported in Rel-16 are listed in Table 2. The separate positioning techniques as indicated in Table 2 may be currently configured and performed based on the requirements of the LMF and/or UE capabilities. Note that Table 2 includes TBS positioning based on PRS signals, but only OTDOA based on LTE signals is supported. The E-CID includes Cell-ID for NR method. The Terrestrial Beacon System ("TBS") method refers to TBS positioning based on Metropolitan Beacon System ("MBS") signals.
(82) TABLE-US-00002 TABLE 2 Supported Rel-16 UE positioning methods UE-assisted, NG-RAN UE- LMF- node Secure User Plane Location Method based based assisted ("SUPL") A-GNSS Yes Yes No Yes (UE-based and UE-assisted) OTDOA No Yes No Yes (UE-assisted) E-CID No Yes Yes Yes for E-UTRA (UE-assisted) Sensor Yes Yes No No WLAN Yes Yes No Yes Bluetooth No Yes No No TBS Yes Yes No Yes (MBS) DL-TDOA Yes Yes No No DL-AoD Yes Yes No No Multi-RTT No Yes Yes No NR E-CID No Yes FFS No UL-TDOA No No Yes No UL-AoA No No Yes No
(83) The transmission of Positioning Reference Signals ("PRS") enables the UE **205** to perform UE positioning-related measurements to enable the computation of a UE's location estimate and are configured per Transmission Reception Point ("TRP"), where a TRP may transmit one or more beams.
(84) FIG. **3** depicts a system **300** for NR beam-based positioning. According to Rel-16, the PRS can be transmitted by different base stations (serving and neighboring) using narrow beams over Frequency Range #1 Between ("FR1", i.e., frequencies from 410 MHz to 7125 MHz) and Frequency Range #2 ("FR2", i.e., frequencies from 24.25 GHz to 52.6 GHz), which is relatively different when compared to LTE where the PRS was transmitted across the whole cell.
(85) As illustrated in FIG. **3**, a UE **205** may receive PRS from a first gNB (″gNB #1) **310** which is a serving gNB, and also from a neighboring second gNB (″gNB #2) **315**, and a neighboring third gNB (″gNB #3) **320**. Here, the PRS can be locally associated with a PRS Resource ID and Resource Set ID for a base station (i.e., TRP). In the depicted embodiments, each gNB **310**, **315**, **320** is configured with a first Resource Set ID **325** and a second Resource Set ID **330**. As depicted, the UE **205** receives PRS on transmission beams; here, receiving PRS from the gNB #1 **310** on PRS Resource ID #1 from the second Resource Set ID **330**, receiving PRS from the gNB #2 **315** on PSR Resource ID #3 from the second Resource Set ID **330**, and receiving PRS from the gNB #3 **320** on PRS Resource ID #3 from the first Resource Set ID **325**.
(86) Similarly, UE positioning measurements such as Reference Signal Time Difference ("RSTD") and PRS RSRP measurements are made between beams as opposed to different cells as was the case in LTE. In addition, there are additional UL positioning methods for the network to exploit in order to compute the target UE's location. Table 3 lists the RS-to-measurements mapping required for each of the supported RAT-dependent positioning techniques at the UE, and Table 4 lists the RS-to-measurements mapping required for each of the supported RAT-dependent positioning techniques at the gNB.
(87) TABLE-US-00003 TABLE 3 UE Measurements to enable RAT-dependent positioning techniques To facilitate support DL/UL Reference of the following Signals UE Measurements positioning techniques Rel-16 DL PRS DL RSTD DL-TDOA Rel-16 DL PRS DL PRS RSRP DL-TDOA, DL-AoD, Multi-RTT Rel-16 DL PRS / UE Rx-Tx time difference Multi-RTT Rel-16 SRS for positioning Rel. 15 SSB / CSI-RS SS-RSRP(RSRP for RRM), E-CID for RRM SS-RSRQ (for RRM), CSI-RSRP (for RRM), CSI-RSRQ (for RRM), SS-RSRPB (for RRM)

(88) TABLE-US-00004 TABLE 4 gNB Measurements to enable RAT-dependent positioning techniques To facilitate support DL/UL of the following Reference Signals gNB Measurements positioning techniques Rel-16 SRS for positioning UL RTOA UL-TDOA Rel-16 SRS for positioning UL SRS-RSRP UL-TDOA, UL-AoA, Multi-RTT Rel-16 SRS for positioning, gNB Rx-Tx time Multi-RTT Rel-16 DL PRS difference Rel-16 SRS for positioning, A-AoA and Z-AoA UL-AoA, Multi-RTT

(89) RAT-dependent positioning techniques involve the 3GPP RAT and core network entities to perform the position estimation of the UE, which are differentiated from RAT-independent positioning techniques which rely on Global Navigation Satellite System ("GNSS"), Inertial Measurement Unit ("IMU") sensor, WLAN and Bluetooth technologies for performing target device (i.e., UE) positioning.

(90) Regarding PRS design, for 3GPP Rel-16, a DL PRS Resource ID in a DL PRS Resource set is associated with a single beam transmitted from a single TRP. Note that a TRP may transmit one or more beams. A DL PRS occasion is one instance of periodically repeated time windows (consecutive slot(s)) where DL PRS is expected to be transmitted. With regards to Quasi Co-Location ("QCL") relations beyond Type-D of a DL PRS resource, support one or more of the following QCL options: QCL Option 1: QCL-TypeC from a Synchronization Signal Block ("SSB") from a TRP. QCL Option 2: QCL-TypeC from a DL PRS resource from a TRP. QCL Option 3: QCL-TypeA from a DL PRS resource from TRP. QCL Option 4: QCL-TypeC from a Channel State Information Reference Signal ("CSI-RS") resource from a TRP. QCL Option 5: QCL-TypeA from a CSI-RS resource from a TRP. QCL Option 6: No QCL relation beyond Type-D is supported.

(91) Note that QCL-TypeA refers to Doppler shift, Doppler spread, average delay, delay spread; QCL-TypeB refers to Doppler shift, Doppler spread'; QCL-TypeC refers to Average delay, Doppler shift; and QCL-TypeD refers to Spatial Rx parameter.

(92) For a DL PRS resource, QCL-TypeC from an SSB from a TRP (QCL Option 1) is supported. An ID is defined that can be associated with multiple DL PRS Resource Sets associated with a single TRP. An ID is defined that can be associated with multiple DL PRS Resource Sets associated with a single TRP. This ID can be used along with a DL PRS Resource Set ID and a DL PRS Resources ID to uniquely identify a DL PRS Resource. Each TRP should only be associated with one such ID.

(93) DL PRS Resource IDs are locally defined within DL PRS Resource Set. DL PRS Resource Set IDs are locally defined within TRP. The time duration spanned by one DL PRS Resource set containing repeated DL PRS Resources should not exceed DL-PRS-Periodicity. Parameter DL-PRS-ResourceRepetitionFactor is configured for a DL PRS Resource Set and controls how many times each DL-PRS Resource is repeated for a single instance of the DL-PRS Resource Set. Supported values include: 1, 2, 4, 6, 8, 16, 32.

(94) As related to NR positioning, the term "positioning frequency layer" refers to a collection of DL PRS Resource Sets across one or more TRPs which have: The same SCS and CP type The same center frequency The same point-A (already agreed) All DL PRS Resources of the DL PRS Resource Set have the same bandwidth All DL PRS Resource Sets belonging to the same Positioning Frequency Layer have the same value of DL PRS Bandwidth and Start PRB

(95) Duration of DL PRS symbols in units of ms a UE can process every T ms assuming 272 PRB allocation is a UE capability.

(96) RRC signaling may be introduced for a UE to request a measurement gap configuration when the UE is expected to measure the DL PRS resource outside the active DL BWP. In case DL PRS Resources are processed in the active BWP and there is no measurement gap configured to the UE, at least in FR2, the UE may not be expected to process DL PRS in the same OFDM symbol where other DL signals and channels are transmitted to the UE. Behavior in FR1 is determined by RAN4.

(97) In one embodiment, configured DL PRS are transmitted on DL symbols of a slot configured by higher layers. In further embodiments, configured DL PRS are transmitted on symbols of slot

configured as flexible symbols by higher layers. In certain embodiments, if the UE is not provided with a measurement gap, the UE is not expected to process DL PRS Resources on serving or neighboring cells on symbols indicated as UL by the serving cell.

(98) In one embodiment, for UE DL PRS processing capability, the UE reports one combination of (N, T) values per band, where N is a duration of DL PRS symbols in ms processed every T ms for a given maximum bandwidth (B) in MHz supported by UE. Additionally, the UE may report new parameters—a number of DL PRS resources that UE can process in a slot, which is reported per SCS per band. The values of which may include 1, 2, 4, 8, 12, 16, 32, 64.

(99) In one embodiment, the following sets of values for N, T and B are supported: N={0.125, 0.25, 0.5, 1, 2, 4, 8, 12, 16, 20, 25, 30, 35, 40, 45, 50} ms, T={8, 16, 20, 30, 40, 80, 160, 320, 640, 1280} ms, and maximum BW reported by UE={5, 10, 20, 40, 50, 80, 100, 200, 400} MHz.

(100) When a UE is configured in the assistance data of a positioning method with a number of PRS resources beyond its capability (FG 13-2,13-3,13-4 for AoD, TDOA, MRTT respectively), the UE assumes the DL-PRS Resources in the assistance data are sorted in a decreasing order of measurement priority. Specifically, in one embodiment, according to the current RAN2 structure of the assistance data, the following priority is assumed: The 4 frequency layers are sorted according to priority; The 64 TRPs per frequency layer are sorted according to priority; The 2 sets per TRP of the frequency layer are sorted according to priority; and The 64 resources of the set per TRP per frequency layer are sorted according to priority.

(101) In one embodiment, the reference indicated by nr-DL-PRS-Referencelnfo-r16 for each frequency layer has the highest priority at least for DL-TDOA.

(102) In some embodiments, the terms antenna, panel, and antenna panel are used interchangeably. An antenna panel may be a hardware that is used for transmitting and/or receiving radio signals at frequencies lower than 6GHz, e.g., frequency range 1 (FR1), or higher than 6GHz, e.g., frequency range 2 (FR2) or millimeter wave (mmWave). In some embodiments, an antenna panel may comprise an array of antenna elements, wherein each antenna element is connected to hardware such as a phase shifter that allows a control module to apply spatial parameters for transmission and/or reception of signals. The resulting radiation pattern may be called a beam, which may or may not be unimodal and may allow the device to amplify signals that are transmitted or received from spatial directions.

(103) In some embodiments, an antenna panel may or may not be virtualized as an antenna port in the specifications. An antenna panel may be connected to a baseband processing module through a radio frequency ("RF") chain for each of transmission (egress) and reception (ingress) directions. A capability of a device in terms of the number of antenna panels, their duplexing capabilities, their beamforming capabilities, and so on, may or may not be transparent to other devices. In some embodiments, capability information may be communicated via signaling or, in some embodiments, capability information may be provided to devices without a need for signaling. In the case that such information is available to other devices, it can be used for signaling or local decision making.

(104) In some embodiments, a device (e.g., UE, node) antenna panel may be a physical or logical antenna array comprising a set of antenna elements or antenna ports that share a common or a significant portion of an RF chain (e.g., in-phase/quadrature ("I/Q") modulator, analog to digital ("A/D") converter, local oscillator, phase shift network). The device antenna panel or "device panel" may be a logical entity with physical device antennas mapped to the logical entity.

(105) The mapping of physical device antennas to the logical entity may be up to device implementation. Communicating (receiving or transmitting) on at least a subset of antenna elements or antenna ports active for radiating energy (also referred to herein as active elements) of an antenna panel requires biasing or powering on of the RF chain which results in current drain or power consumption in the device associated with the antenna panel (including power amplifier/low noise amplifier ("LNA") power consumption associated with the antenna elements or antenna

ports). The phrase “active for radiating energy,” as used herein, is not meant to be limited to a transmit function but also encompasses a receive function. Accordingly, an antenna element that is active for radiating energy may be coupled to a transmitter to transmit radio frequency energy or to a receiver to receive radio frequency energy, either simultaneously or sequentially, or may be coupled to a transceiver in general, for performing its intended functionality. Communicating on the active elements of an antenna panel enables generation of radiation patterns or beams.
(106) In some embodiments, depending on device's own implementation, a “device panel” can have at least one of the following functionalities as an operational role of Unit of antenna group to control its Tx beam independently, Unit of antenna group to control its transmission power independently, Unit of antenna group to control its transmission timing independently. The “device panel” may be transparent to the RAN node. For certain condition(s), the RAN node **210** can assume the mapping between device's physical antennas to the logical entity “device panel” may not be changed. For example, the condition may include until the next update or report from device or comprise a duration of time over which the RAN node assumes there will be no change to the mapping.
(107) A device may report its capability with respect to the “device panel” to the RAN node or network. The device capability may include at least the number of “device panels.” In one implementation, the device may support UL transmission from one beam within a panel; with multiple panels, more than one beam (one beam per panel) may be used for UL transmission. In another implementation, more than one beam per panel may be supported/used for UL transmission.
(108) In some of the embodiments described, an antenna port is defined such that the channel over which a symbol on the antenna port is conveyed can be inferred from the channel over which another symbol on the same antenna port is conveyed.
(109) Two antenna ports are said to be quasi co-located if the large-scale properties of the channel over which a symbol on one antenna port is conveyed can be inferred from the channel over which a symbol on the other antenna port is conveyed. The large-scale properties include one or more of delay spread, Doppler spread, Doppler shift, average gain, average delay, and spatial Rx parameters.
(110) Two antenna ports may be quasi-located with respect to a subset of the large-scale properties and different subset of large-scale properties may be indicated by a Quasi-Co-Location (“QCL”) Type. For example, the parameter qcl-Type may take one of the following values: ‘QCL-TypeA’: {Doppler shift, Doppler spread, average delay, delay spread} ‘QCL-TypeB’: {Doppler shift, Doppler spread} ‘QCL-TypeC’: {Doppler shift, average delay} ‘QCL-TypeD’: {Spatial Rx parameter}.
(111) Spatial Rx parameters may include one or more of: Angle of Arrival (“AoA”), Dominant AoA, average AoA, angular spread, Power Angular Spectrum (“PAS”) of AoA, average Angle of Departure (“AoD”), PAS of AoD, transmit/receive channel correlation, transmit/receive beamforming, spatial channel correlation etc.
(112) An “antenna port” according to an embodiment may be a logical port that may correspond to a beam (resulting from beamforming) or may correspond to a physical antenna on a device. In some embodiments, a physical antenna may map directly to a single antenna port, in which an antenna port corresponds to an actual physical antenna. Alternately, a set or subset of physical antennas, or antenna set or antenna array or antenna sub-array, may be mapped to one or more antenna ports after applying complex weights, a cyclic delay, or both to the signal on each physical antenna. The physical antenna set may have antennas from a single module or panel or from multiple modules or panels. The weights may be fixed as in an antenna virtualization scheme, such as cyclic delay diversity (“CDD”). The procedure used to derive antenna ports from physical antennas may be specific to a device implementation and transparent to other devices.
(113) In some of the embodiments described, a TCI-state associated with a target transmission can

indicate parameters for configuring a quasi-collocation relationship between the target transmission (e.g., target RS of DM-RS ports of the target transmission during a transmission occasion) and a source reference signal(s) (e.g., SSB/CSI-RS/SRS) with respect to QCL type parameter(s) indicated in the corresponding TCI state. A device can receive a configuration of a plurality of transmission configuration indicator states for a serving cell for transmissions on the serving cell.
(114) In some of the embodiments described, a spatial relation information associated with a target transmission can indicate parameters for configuring a spatial setting between the target transmission and a reference RS (e.g., SSB/CSI-RS/SRS). For example, the device may transmit the target transmission with the same spatial domain filter/beam used for reception the reference RS (e.g., DL RS such as SSB/CSI-RS). In another example, the device may transmit the target transmission with the same spatial domain transmission filter/beam used for the transmission of the reference RS (e.g., UL RS such as SRS). A device can receive a configuration of a plurality of spatial relation information configurations for a serving cell for transmissions on the serving cell.
(115) Regarding physical layer latency, start and end times may be defined as shown in Table 5 below:
(116) TABLE-US-00005 TABLE 5 Physical Layer Latency Start and End times Method Start End UE assisted Transmission of the PDSCH Successful decoding of the DL-only & from the gNB carrying the PUSCH carrying the LPP DL-ECID & LPP Request Location Provide Location Multi-RTT Information message Information message UL-only Reception by the gNB of the The transmission by method & NRPPa measurement request the gNB of the NRPPa UL ECID & message measurement response Multi-RTT message UE-based Transmission of the PDSCH Successful decoding of the from the gNB carrying the PUSCH at gNB carrying the LPP Request Location LPP Provide Location Information if applicable, Information message if otherwise, applicable, otherwise Alt. 1: transmission of Calculation of Location the PUSCH carrying the Estimate at the UE MG Request from the UE. Alt. 2: Transmission of the PDSCH from the gNB carrying the LPP message containing the assistance data Alt. 3: Start of the Reception of DL PRS Note: Suggest to downselect this at the next meeting. Note: The high layers latency components may be subject to adjustment for different alternatives.
(117) In one embodiment, physical layer latency for DL only, UL only, DL+UL positioning solutions for UE-based and UE-assisted approaches are separately defined. In certain embodiments, at least the following information is provided for positioning physical layer latency analysis: Source initiating request for positioning measurements/location for a given UE (UE, Network) Destination awaiting for positioning measurements/location for a given UE (UE, Network) Start and end triggers/events for physical layer latency evaluation, which, for Rel.16 solutions, is based on specification for each solution Initial and final RRC State of positioned UE (RRC IDLE, INACTIVE, CONNECTED) at the start and end time for the physical layer latency evaluation Positioning a. technique (enumeration): (1) DL-TDOA, (2) DL AoD, (3) UL-TDoA, (4) UL-AoA, (5) Multi-RTT, (6) E-CID b. type: DL, UL, DL+UL c. mode: UE-based, UE-assisted Latency component w/value range and description, including information on any parallel (simultaneous) components Total latency value
(118) In one embodiment, semi-persistent and aperiodic transmission, and reception of DL PRS is used, which may include UE-assisted and/or UE-based positioning and DL positioning and/or Multi-RTT.
(119) In one embodiment, on-demand transmission, and reception of DL PRS is used, which may include UE-assisted and/or UE-based positioning and DL positioning and/or Multi-RTT. As used herein, semi-persistent means MAC-CE triggered, aperiodic corresponds to DCI-triggered, and on-demand corresponds to the UE-initiated or network-initiated request of PRS and/or SRS. Thus, in one embodiment, it is not the same as whether PRS is DCI-triggered or MAC-CE triggered; rather, it is about UE or LM requesting/suggesting/recommending specific PRS patterns, ON/OFF, periodicity, BW, and/or the like.

(120) Regarding RAN4 positioning, in one embodiment, Rel-15 measurement gap ("MG") patterns are applicable for positioning measurements. If new MG patterns are introduced, the new MG patterns are UE capability. In one embodiment, the handling of LTE PRS in Rel-15 CSSF for gap sharing between NR PRS and RRM is re-used.
(121) In some embodiments regarding the PRS measurement period when incomplete PRS measurement in active BWP is abandoned and restarted in gaps, additional requirements are not defined, but capture the above UE behavior in the relevant requirements for positioning measurement being performed within the active BWP.
(122) In certain embodiments, for a UE that does not need any PRS and/or RRM measurement relaxation due to concurrent processing of PRS and RRM measurements, and for a UE that needs PRS and/or RRM measurement relaxation due to concurrent processing of PRS and RRM measurements, UE capability signaling may indicate that the concurrent processing of PRS and RRM measurements does not need any PRS and/or RRM measurement relaxation.
(123) In addition to Rel-15 measurement gap patterns, RAN4 introduces in Rel-16 new measurement gap patterns applicable for UEs configured with NR positioning measurements, including that the number of the new measurement gap patterns is two and the new measurement gap patterns are UE capability.
(124) In one embodiment, UEs may support the measurement gap patterns listed in Table 6. The UE may determine measurement gap timing based on a gap offset configuration and measurement gap timing advance configuration provided by higher layer signaling.
(125) TABLE-US-00006 TABLE 6 Measurement Gap Pattern Configurations Measurement Gap Measurement Gap Repetition Gap Pattern Id Length (MGL, ms) Period (MGRP, ms)   0 6 40   1 6 80   2 3 40   3 3 80   4 6 20   5 6 160   6 4 20   7 4 40   8 4 80   9 4 160 10 3 20 11 3 160 12 5.5 20 13 5.5 40 14 5.5 80 15 5.5 160 16 3.5 20 17 3.5 40 18 3.5 80 19 3.5 160 20 1.5 20 21 1.5 40 22 1.5 80 23 1.5 160 24 10 80 25 20 160
(126) In one embodiment, regarding measurement and report configuration, UE measurements have been defined, which are applicable to DL-based positioning techniques.
(127) FIG. **4** depicts one example of an Information Element **400**, i.e., NR-DL-TDOA-ProvideAssistanceData, used by the location server to provide an assistance data configuration to enable UE-assisted and UE-based NR downlink TDOA. The depicted Information Element ("IE") may also be used to provide NR DL TDOA positioning specific error reason.
(128) FIG. **5** shows one example of an Information Element **500**, i.e., NR-DL-TDOA-SignalMeasurementInformation, used by the target device (i.e., UE **205**) to provide NR-DL TDOA measurements to the location server. The measurements are provided as a list of TRPs, where the first TRP in the list is used as reference TRP in case RSTD measurements are reported. The first TRP in the list may or may not be the reference TRP indicated in the NR-DL-PRS-AssistanceData. Furthermore, the target device selects a reference resource per TRP, and compiles the measurements per TRP based on the selected reference resource.
(129) Regarding RAT-dependent positioning measurements, the different DL measurements including DL PRS-RSRP, DL RSTD and UE Rx-Tx Time Difference required for the supported RAT-dependent positioning techniques. The following measurement configurations are specified: 4 Pair of DL RSTD measurements can be performed per pair of cells. Each measurement is performed between a different pair of DL PRS Resources/Resource Sets with a single reference timing. 8 DL PRS RSRP measurements can be performed on different DL PRS resources from the same cell.
(130) TABLE-US-00007 TABLE 7 DL Measurements required for DL-based positioning methods DL PRS reference signal received power (DL PRS-RSRP) Definition DL PRS reference signal received power (DL PRS-RSRP), is defined as the linear average over the power contributions (in [W]) of the resource elements that carry DL PRS reference signals configured for RSRP measurements within the considered measurement frequency bandwidth. For frequency range 1,

the reference point for the DL PRS-RSRP shall be the antenna connector of the UE. For frequency range 2, DL PRS-RSRP shall be measured based on the combined signal from antenna elements corresponding to a given receiver branch. For frequency range 1 and 2, if receiver diversity is in use by the UE, the reported DL PRS-RSRP value shall not be lower than the corresponding DL PRS-RSRP of any of the individual receiver branches. Applicable RRC_CONNECTED intra-frequency, for RRC CONNECTED inter-frequency DL reference signal time difference (DL RSTD) Definition DL reference signal time difference (DL RSTD) is the DL relative timing difference between the positioning node j and the reference positioning node i, defined as T.sub.SubframeRxj-T.sub.SubframeRxi, Where: T.sub.SubframeRxj is the time when the UE receives the start of one subframe from positioning node j. T.sub.SubframeRxi is the time when the UE receives the corresponding start of one subframe from positioning node i that is closest in time to the subframe received from positioning node j. Multiple DL PRS resources can be used to determine the start of one subframe from a positioning node. For frequency range 1, the reference point for the DL RSTD shall be the antenna connector of the UE. For frequency range 2, the reference point for the DL RSTD shall be the antenna of the UE. Applicable RRC_CONNECTED intra-frequency for RRC_CONNECTED inter-frequency UE Rx-Tx time difference Definition The UE Rx-Tx time difference is defined as T.sub.UE-RX-T.sub.UE-TX Where: T.sub.UE-RX is the UE received timing of downlink subframe #i from a positioning node, defined by the first detected path in time. T.sub.UE-TX is the UE transmit timing of uplink subframe #j that is closest in time to the subframe #i received from the positioning node. Multiple DL PRS resources can be used to determine the start of one subframe of the first arrival path of the positioning node. For frequency range 1, the reference point for T.sub.UE-RX measurement shall be the Rx antenna connector of the UE and the reference point for T.sub.UE-TX measurement shall be the Tx antenna connector of the UE. For frequency range 2, the reference point for T.sub.UE-RX measurement shall be the Rx antenna of the UE and the reference point for T.sub.UE-TX measurement shall be the Tx antenna of the UE. Applicable RRC_CONNECTED intra-frequency for RRC_CONNECTED inter-frequency
(131) The present embodiments include techniques to enable configurations related to a UE's positioning processing capabilities and UL resource availability to enable positioning in a variety of latency and accuracy scenarios, including low latency and high accuracy positioning. Note that these embodiments can be used in combination with one another depending on the implementation.
(132) In a first embodiment, the UE processing timeline of DL-based positioning methods is discussed, which requires measurements pertaining to the DL-PRS in order to obtain the target-UE's location estimate, e.g., RSTD, UE Rx-Tx Time Difference, DL-PRS RSRP. In this aspect, the presented solutions are tailored towards UE-assisted (location estimate is computed at the LMF) and UE-based (location estimate is locally computed at the UE) positioning.
(133) In one embodiment, UE-assisted positioning involves signaling exchanges among the serving gNB, target-UE and finally LMF, in determining the location estimate. FIG. **6** illustrates the systematic procedures for target-UE performing positioning in a scenario where the PRS is processed within a DL BWP.
(134) The UE processing timeline is measured from the time instance that the target-UE receives the DL-PRS physical layer configuration to the time instance that the target-UE transmits the measurement report to the serving gNB. It can be noted that in FIG. **6**: Y parameter **602** defines the duration between the UE receiving the DL-PRS configuration in the ProvideAssistanceData message and receiving the RequestLocationInformation message containing the measurement configuration including measurements and number to be reported. X parameter **604** defines the duration between the UE receiving the RequestLocationInformation message and the UE transmitting the ProvideLocationInformation message containing the measurement report.
(135) In one embodiment, the Y duration **602** mainly depends on when the UE receives the DL-PRS configuration (e.g., via broadcast or dedicated signaling), which can occur when the UE is in either the RRC_IDLE/INACTIVE or RRC_CONNECTED state. Depending on the time instance

when Step (2) has been triggered, the target-UE may store the DL-PRS configuration for period defined by Y **602**, which can vary depending on the target-UE state.
(136) The X duration **604** is dependent on the configured positioning method by the LMF and number of measurements to be performed, as noted in Table 8. Table 8 further indicates the maximum number of supported measurements per target-UE. The X duration **604** is not limited to the techniques indicated in Table 8 but may also correspond to any positioning method and corresponding measurements configured by the location server.
(137) TABLE-US-00008 TABLE 8 Maximum number of measurements per configured positioning technique DL-based Maximum Positioning Number of Technique Measurement Report Type Measurements DL-AoD DL-PRS RSRP measurements 8 on different PRS resources from the same TRP supported by the UE DL-TDOA DL-PRS RSTD Measurement Report 4 DL-PRS RSRP Measurement 8 Multi-RTT UE Rx-Tx Measurement Report 4 DL-PRS RSRP Measurement 8
(138) In an alternate embodiment, the UE receives the DL-PRS configuration and corresponding reporting configuration together. In this scenario, a combined processing timeline could be assumed that includes processing the configuration, performing the DL-PRS measurements, and processing the report.
(139) The LMF **144** can configure a set of X **604** (and Y **602**) values for the UE depending on the following factors: UE capability of the UE: Reduced capability positioning UEs, will have relaxed timing requirements when compared to UEs with enhanced capabilities. Positioning Latency Budget: The positioning service would have a relaxed to stringent Time-to-Find-First-Fix (“TTFF”). Accuracy requirement: Depending on the number of measurements to be performed within the X time window, the positioning accuracy may be low or high.
(140) The X and Y values **602**, **604** can depend on the required positioning latency budget required by an LCS client or application function. The UE processing configuration can be signaled in at least one of the following methods: Via dedicated signaling for UE-specific processing timeline configurations, depending on e.g., latency budget of the positioning service. a. Relaxed latency requirements may use LPP signaling b. Stringent latency requirements may use dynamic L1/L2 signaling such as DCI/MAC CE/RRC signaling Via system information broadcast signaling, e.g., SIB/on-demand signaling for a set of UEs with the shared aforementioned criteria.
(141) In one example implementation, when the UE is configured to report measurement quantities related to DL-TDOA, then depending up on the UE processing capability, two embodiments can be considered: In one embodiment, the UE is able to process both DL-PRS RSTD measurements and DL-PRS RSRP measurements at the same time such that the processing timeline X is comprised of a single value. In another embodiment, the UE is able to process DL-PRS RSTD measurements and DL-PRS RSRP measurements in a sequential manner such that the processing timeline X can be composed of two timelines X1 and X2, respectively.
(142) FIG. **7** illustrates the procedures related to UE-based positioning and the corresponding UE processing timeline associated with such procedures. Similar to the embodiment shown in FIG. **6**, the DL-PRS is also processed within a DL-BWP.
(143) In the case of UE-based positioning, in one embodiment, the target-UE must initiate Step 1 if there is: no prior DL-PRS physical layer configuration stored in the UE, existing DL-PRS configuration is outdated, or if the existing DL-PRS configuration does not match the accuracy requirements.
(144) Duration Z **702** between Steps (1) **701** and (2) **703**, in one embodiment, depends on the scheduling latency of the LMF to provide the desired measurement configuration. Steps (2) **703** and (3) **705** are similar to the embodiment depicted in FIG. **6**, in that the UE processing delay is dependent on the duration between the instance that the UE receives the DL-PRS physical layer configuration and the instance that the required number of measurements have been gathered (given by U **704**). V **706** is the processing duration to compute the location estimate at the target-UE. Steps (5) **707** and (6) **709** are optionally required if the LMF **144** would like the target-UE's

location to estimate to be reported and therefore may not affect the target-UE's positioning processing timeline, unlike in the embodiment shown in FIG. **6** where Step (3) has a direct impact on the UE's processing timeline. Similarly, in Step (4) **711**, the target-UE may locally compute a location estimate based on the positioning measurements.
(145) In further embodiments, the measurement of a DL-PRS configuration extends to outside an active DL BWP of a serving cell and requires DL-PRS measurements of a TRP from a neighboring cell for enhanced accuracy positioning. However, in order to measure the DL-PRS resources outside an DL BWP of a serving cell/frequency, in one embodiment, a measurement gap would have to be configured at the target-UE, which can be provided to the UE or be provided upon request. This may add to the UE processing load and delay. At present, there are few issues in Rel-16 positioning to be considered with respect to the UE positioning processing capability: RRM and Positioning Measurement Gap configuration are shared and thus the time-frequency location of the DL-PRS resources must be within the same SMTC window as the SSBs of the corresponding non-serving cell to be measured. The UE processing timeline is impacted by the Measurement Gap Length ("MGL") and may not be optimized for reducing positioning latency. There is a delay associated with the target-UE receiving the measurement gap ("MG") configuration (e.g., via RRC) and subsequent application of this configuration.
(146) FIG. **8** illustrates the impact of the MG configuration on the UE positioning processing timeline, which is associated with an MGL **803** and a Measurement Gap Repetition Period ("MGRP") **801**. The delay associated with requesting and/or receiving an MG configuration is not shown.
(147) In the case of both RRM and positioning, in one embodiment, a per-UE or per-FR MG can be configured. This implies that the MGL **803** should accommodate an SMTC and PRS occasion. A trade-off exists between the UE processing load and the length and periodicity of MGRP **801** to accommodate PRS occasions to be measured for high accuracy positioning. The RF tuning time **807** at the start and end of the MGL is also shown to contribute to the MGL **803**.
(148) The depicted embodiment presents a hybrid MG configuration for positioning where {X1, X2, . . . , XN} **805** can be adapted based on the set of criteria, as mentioned in the embodiment depicted in FIG. **6**: Capability of the UE: Reduced capability positioning UEs, will have relaxed timing requirements when compared to UEs with enhanced capabilities. Positioning Latency Budget: The positioning service would have a relaxed to stringent Time-to-Find-First-Fix ("TTFF"). Accuracy requirement: Depending on the number of measurements to be performed within the X time window, the positioning accuracy may be low or high.
(149) In further embodiments, for a target UE, PRS processing unit ("PPU") is proposed such that the processing capability for that UE could be defined in terms of number of PPUs it could support for a given symbol to process the PRS measurements and reports. Furthermore, for each type of PRS measurement and reporting, UE capability may be defined in terms of number of PPUs required to process corresponding report.
(150) In one example implementation of this embodiment, when a UE is configured to report measurements for DL-TDOA and the UE is capable of M PPUs, then if DL-PRS RSTD requires N PPUs in a symbol, the remaining M-N PPUs may be used for DL-PRS RSRP, if it is sufficient, otherwise no parallel processing is possible on the same symbol. In such an embodiment, sequential processing may be performed.
(151) In one embodiment, a mechanism for the UE to provision UL resources for transmitting positioning measurement report(s) within a defined period to the LMF 144 is described. Currently NR defines two types of the configured grants, viz. Type 1 and Type 2 grants: Type 1 grants can be configured via RRC including the periodicity Type 2 grants can be activated/deactivated via DCI.
(152) In the case of UE-assisted positioning, the measurement report may be transmitted via the ProvideLocation message (higher layer NAS signaling), which is inherently non-dynamic when compared to the Type 1 and Type 2 configured grants, which are based on L1/L2 signaling For UE-

based positioning, the ProvideLocation message provides the computed location estimate of the UE.

(153) The LMF **144** may request the serving gNB to configure UL grants once the positioning-related measurements, e.g., based on prior DL-PRS transmissions, are ready to be reported (e.g., Step 4 of FIG. 6 or Step 6 of FIG. **7**). FIG. **9** is an illustration of the dynamic reporting mechanism. In FIG. **9**(*a*), Step (1) **901**, the UE receives the UL CG configuration, containing exemplary configuration details such as the time-frequency resources, activation indication, offset, and/or periodicity. The serving gNB may have prior confirmation through message exchanges with the LMF **144** regarding the scheduling of the DL-PRS with other neighboring cells since a target-UE of a serving cell can only be configured with a UL Type 1 activation (FIG. **9**(*a*)) or a Type 2 activation (FIG. **9**(*b*)). Steps (2)-(5) **903-909** of FIG. **9**(*a*) include the transmission of the positioning report based on earliest availability.

(154) In an alternative implementation, the measurements may be ranked according to measurement priority or as per the positioning latency budget and transmitted accordingly. In FIG. **9**(*b*), the UL CG may be deactivated, at Step (6) **913**, using explicit signaling e.g., using the ProvideLocation message. In another implementation example, the ProvideLocation message may also contain another UL CG activation for the next UL configured grant for measurement reporting.

(155) In another implementation, explicit indication of the deactivation of the UL CG with activation message (after a certain configured time) can be indicated in Step (1) **911**, when the UE first receives the UL CG configuration.

(156) In an alternate embodiment, when a UE is configured with a PRS measurement report that may contain multiple quantities to be reported for the corresponding positioning techniques, then partial reporting could be done (specially for low-latency requirements), where multiple UL resources are configured or indicated, and partial reporting is done on different instances of the UL resources. Basically, in one embodiment, for a partial report, instead of processing the entire report, the UE starts reporting the individual parts as they are ready. The exact sequence of partial reporting, e.g., which quantity is reported earlier than the other may be configured to the UE explicitly to implicitly based on the required processing timeline for each quantity.

(157) In further embodiments, a method of prioritization of PRS measurement reports based on availability of UL resources is described. In the scenario where the is a limited availability of UL CG resources for transmitting all the readily available measurement reports, in certain embodiments, a prioritization criteria may be applied to each of the positioning measurements based on a certain criteria such as positioning latency budget, accuracy, and type of positioning method.

(158) The prioritization criteria may be configured by the LMF **144** via e.g., a ProvideAssistanceData message in the case of UE-assisted positioning methods. In the case of UE-based positioning methods, the UE may indicate to the LMF **144** and/or gNB, the preferred criteria of the requested message via e.g., RequestAssistanceData message on the PUSCH. This enables efficient processing of the DL-PRS based on the associated priority criteria of each of the measurements.

(159) In an alternative implementation, the target-UE may request the preferred priority of the positioning-related reference signal measurements, e.g., DL-PRS, SRS, in an on-demand manner using either L1, e.g., DCI, or L2 e.g., RRC/MAC CE signaling.

(160) In one example, when the UE is required to process multiple PRS reports and each report is assigned or indicated with a priority level, then the UE starts with the highest priority report and calculates the available processing units and assigns the required processing units to 1.sup.st report with highest priority. Then the UE checks the remaining processing units and the required processing units for a 2.sup.nd report with second highest priority and if the remaining available units are sufficient, then the UE is capable of parallel processing the 2.sup.nd report as well. The UE then continues this process until it runs out of sufficient processing units. In that case, the UE

can either delay the processing of lower priority reports, if the latency requirement could still be met. Otherwise, the UE may drop the low priority reports that cannot be processed within the required latency constraint.

(161) In another example, the UE is required to perform measurements from multiple TRPs and process corresponding reports. If the priority associated or the associated accuracy is lower compared to other configured reports, then depending upon the availability of processing units, the UE could process all the measurements from all the TRPs in parallel or in a sequential manner with some delay. However, if such delays exceed the required latency constraint, then the UE drops measurements (or measurement reports) from one or more TRPs and only reports partial measurements (or measurement reports) from a sub-set of TRPs.

(162) There may be cases where incomplete measurements may arise during a positioning measurement window resulting in an incomplete report. In order to increase the signal efficiency of positioning reporting by a target-UE, in one embodiment, the target-UE may be configured to drop measurements based on certain criteria including: If a measurement report size based on a positioning technique exceeds the UL transmissions resource availability. If measurements are lower in priority with respect to other high priority measurements. If measurements are incomplete or corrupted, e.g., due to failure events and thus the report is not deemed beneficial for processing by the location server (LMF). If measurements are not deemed to be reliable and/or do not satisfy the integrity requirements such as: a. Target Integrity Risk ("TIR"); b. Alert Limit ("AL"); c. Time-to-Alert ("TTA"); d. Protection Level ("PL").

(163) The TIR may be further defined as the probability that the positioning error exceeds the AL without warning the user within the required TTA. The AL, as used herein, is defined as the maximum allowable positioning error such that the positioning system is available for the intended application. If the positioning error is beyond the AL, operations may be hazardous, and the positioning system should be declared unavailable for the intended application to prevent loss of integrity. The TTA, as used herein, is referred to as the maximum allowable elapsed time from when the positioning error exceeds the AL until the function providing position integrity annunciates a corresponding alert. The PL, as used herein, is a statistical upper-bound of the positioning error that ensures that the probability per unit of time of the true error being greater than the AL and the PL being less than or equal to the AL, for longer than the TTA, is less than the required TIR.

(164) In one embodiment, the target-UE may explicitly indicate the dropped measurements or the LMF **144** may implicitly infer the dropped measurements based on the provided measurement configuration.

(165) FIG. **10** depicts a user equipment apparatus **1000** that may be used for configuring positioning measurements and reports, according to embodiments of the disclosure. In various embodiments, the user equipment apparatus **1000** is used to implement one or more of the solutions described above. The user equipment apparatus **1000** may be one embodiment of the remote unit **105** and/or the UE **205**, described above. Furthermore, the user equipment apparatus **1000** may include a processor **1005**, a memory **1010**, an input device **1015**, an output device **1020**, and a transceiver **1025**.

(166) In some embodiments, the input device **1015** and the output device **1020** are combined into a single device, such as a touchscreen. In certain embodiments, the user equipment apparatus **1000** may not include any input device **1015** and/or output device **1020**. In various embodiments, the user equipment apparatus **1000** may include one or more of: the processor **1005**, the memory **1010**, and the transceiver **1025**, and may not include the input device **1015** and/or the output device **1020**.

(167) As depicted, the transceiver **1025** includes at least one transmitter **1030** and at least one receiver **1035**. In some embodiments, the transceiver **1025** communicates with one or more cells (or wireless coverage areas) supported by one or more base units **121**. In various embodiments, the transceiver **1025** is operable on unlicensed spectrum. Moreover, the transceiver **1025** may include

multiple UE panels supporting one or more beams. Additionally, the transceiver **1025** may support at least one network interface **1040** and/or application interface **1045**. The application interface(s) **1045** may support one or more APIs. The network interface(s) **1040** may support 3GPP reference points, such as Uu, N1, PC5, etc. Other network interfaces **1040** may be supported, as understood by one of ordinary skill in the art.

(168) The processor **1005**, in one embodiment, may include any known controller capable of executing computer-readable instructions and/or capable of performing logical operations. For example, the processor **1005** may be a microcontroller, a microprocessor, a central processing unit ("CPU"), a graphics processing unit ("GPU"), an auxiliary processing unit, a field programmable gate array ("FPGA"), or similar programmable controller. In some embodiments, the processor **1005** executes instructions stored in the memory **1010** to perform the methods and routines described herein. The processor **1005** is communicatively coupled to the memory **1010**, the input device **1015**, the output device **1020**, and the transceiver **1025**.

(169) In various embodiments, the processor **1005** controls the user equipment apparatus **1000** to implement the above described UE behaviors. In certain embodiments, the processor **1005** may include an application processor (also known as "main processor") which manages application-domain and operating system ("OS") functions and a baseband processor (also known as "baseband radio processor") which manages radio functions.

(170) In one embodiment, the **1025** transceiver receives, from a mobile wireless communication network, a positioning configuration defining a positioning configuration timeline and a measurement and processing time window for the UE. The positioning configuration may include a timeline duration defining when to start performing measurements, a set of positioning measurements to be taken within the configured time window, and a window duration for measuring and processing requested position-relation measurements for the UE according to the positioning processing timeline.

(171) In one embodiment, the processor **1005** performs at least one positioning measurement for the UE according to the positioning processing timeline in response to receiving the positioning configuration. In certain embodiments, the transceiver **1025** sends a positioning measurement report comprising the at least one positioning measurement and measurement timeline performed of the at least one positioning measurement within the configured time window from the UE to the mobile wireless communication network.

(172) In one embodiment, multiple one of configuration timelines and measurement and processing timelines may be configured for the UE, and wherein a latency for at least one of a reference signal received power ("RSRP"), a reference signal time difference ("RSTD"), and a UE Rx-Tx positioning measurement within each of the multiple measurement and processing timelines may be reported to the mobile wireless communication network.

(173) In one embodiment, the latency may comprise a single value in response to positioning measurements being performed at a same time and multiple values in response to positioning measurements being performed sequentially.

(174) In one embodiment, the transceiver **1025** receives pre-configured assistance data for positioning from the mobile wireless communication network during a Long-term evolution Protocol Positioning ("LPP") session and performs measurements and processing on the pre-configured assistance data in response to the transceiver receiving an LPP Request Location Information message.

(175) In one embodiment, the transceiver **1025** receives the positioning configuration from the mobile wireless communication network via a broadcast signal, the positioning configuration designed for a plurality of UEs with same capabilities.

(176) In one embodiment, the transceiver **1025** receives the positioning configuration from the mobile wireless communication network via a UE-specific dedicated signal, the UE-specific dedicated signal comprising an LPP Request Location Information message.

(177) In one embodiment, in response to the UE initiating a positioning reference signal ("PRS") configuration request, the processor **1005** determines an overall timeline between receiving the configuration request and receiving the UE's position estimate, the overall timeline comprising multiple timelines related to the configuration of assistance data, measurement, processing, and calculation of the UE's position estimate.
(178) In one embodiment, the transceiver **1025** sends an on-demand request to the mobile wireless communication network to receive downlink-PRS ("DL-PRS") assistance data in response to at least one of no prior DL-PRS physical layer configuration stored at the UE, existing DL-PRS configuration is outdated, and the existing DL-PRS configuration does not meet accuracy requirements.
(179) In one embodiment, the positioning configuration further comprises a set of measurement gap configurations to be applied by the UE for the positioning processing timeline, the measurement gap configuration defining a measurement gap length and a measurement gap length and a measurement gap repetition period for the positioning processing timeline.
(180) In one embodiment, the set of measurement gap configurations can be pre-configured in the UE via signaling from the mobile wireless communication network. In one embodiment, the processor **1005** processes PRS measurements and reports according to a UE PRS processing unit ("PPU"), the UE PPU comprising a number of PPUs that the UE can support for a given symbol.
(181) In one embodiment, based on the UE's capabilities, the processor **1005** performs parallel processing of the same PRS symbol with respect to other positioning measurements. In one embodiment, the mobile wireless communication network comprises at least one of a base station and a location management function.
(182) In further embodiments, the transceiver **1025** receives, from a mobile wireless communication network, an uplink ("UL") configured grant configuration based on criteria associated with at least one of a measurement priority order, a positioning latency budget and a positioning processing timeline for the UE. In some embodiments, the processor **1005** performs at least one positioning measurement for the UE according to at least one of the measurement priority order and the positioning processing timeline and generates a positioning measurement report comprising the at least one positioning measurement.
(183) In certain embodiments, the transceiver **1025** sends the positioning measurement report to the mobile wireless communication network using the UL configured grant configuration based on an availability of positioning-related reference signal measurements based on at least one of the measurement priority order and the positioning processing timeline.
(184) In one embodiment, the UE is configured with a Type 1 UL configured grant configuration via radio resource control ("RRC") signaling for sending the positioning measurement report comprising the at least one positioning measurement.
(185) In one embodiment, the UE is configured with a Type 2 UL configured grant configuration via downlink control information ("DCI") signaling for sending the positioning measurement report comprising the at least one positioning measurement.
(186) In one embodiment, the UL configured grant configuration comprises signaling information for one or more of an offset, a periodicity, an activation, a deactivation, and time-frequency resources of the positioning measurement report comprising the at least one positioning measurement.
(187) In one embodiment, the transceiver **1025** receives a positioning measurement configuration indicating a positioning measurement priority order based on the availability of positioning-related reference signal resources, the priority order determined based on at least one of the UE's capabilities, the positioning latency budget, and a location estimate accuracy. In one embodiment, the processor **1005** performs the at least one positioning measurement and generates the positioning measurement report according to the priority order indicated in the positioning measurement configuration. In one embodiment, the transceiver **1025** sends the positioning measurement report

to the mobile wireless communication network using the UL configured grant configuration according to the priority order indicated in the measurement configuration.

(188) In one embodiment, the transceiver **1025** receives the positioning measurement configuration in response to sending a request for the positioning measurement configuration. In one embodiment, the positioning measurement priority order is configured by at least one of a location server and a base station of the mobile wireless communication network. In one embodiment, the location server exchanges information with the base station related to the configuration and scheduling of the physical uplink shared channel ("PUSCH").

(189) In one embodiment, the transceiver **1025** dynamically provides the positioning measurement configuration, on-demand, using at least one of radio resource control ("RRC") signaling and medium access control ("MAC") control element ("CE") signaling. In one embodiment, the processor **1005** drops the positioning-related reference signal measurements to be reported in response to at least one of a measurement report size exceeding UL transmission resource availability, measurements being lower in priority with respect to other measurements with higher priority, measurements being one of incomplete and corrupted, and measurements determined to be unreliable in response to not satisfying one or more integrity requirements.

(190) In one embodiment, the processor **1005** drops the positioning-related reference signal measurements based on at least one of the latency budget and the measurement and processing timeline.

(191) The memory **1010**, in one embodiment, is a computer readable storage medium. In some embodiments, the memory **1010** includes volatile computer storage media. For example, the memory **1010** may include a RAM, including dynamic RAM ("DRAM"), synchronous dynamic RAM ("SDRAM"), and/or static RAM ("SRAM"). In some embodiments, the memory **1010** includes non-volatile computer storage media. For example, the memory **1010** may include a hard disk drive, a flash memory, or any other suitable non-volatile computer storage device. In some embodiments, the memory **1010** includes both volatile and non-volatile computer storage media.

(192) In some embodiments, the memory **1010** stores data related to configuring positioning measurements and reports. For example, the memory **1010** may store various parameters, panel/beam configurations, resource assignments, policies, and the like as described above. In certain embodiments, the memory **1010** also stores program code and related data, such as an operating system or other controller algorithms operating on the apparatus **1000**.

(193) The input device **1015**, in one embodiment, may include any known computer input device including a touch panel, a button, a keyboard, a stylus, a microphone, or the like. In some embodiments, the input device **1015** may be integrated with the output device **1020**, for example, as a touchscreen or similar touch-sensitive display. In some embodiments, the input device **1015** includes a touchscreen such that text may be input using a virtual keyboard displayed on the touchscreen and/or by handwriting on the touchscreen. In some embodiments, the input device **1015** includes two or more different devices, such as a keyboard and a touch panel.

(194) The output device **1020**, in one embodiment, is designed to output visual, audible, and/or haptic signals. In some embodiments, the output device **1020** includes an electronically controllable display or display device capable of outputting visual data to a user. For example, the output device **1020** may include, but is not limited to, a Liquid Crystal Display ("LCD"), a Light-Emitting Diode ("LED") display, an Organic LED ("OLED") display, a projector, or similar display device capable of outputting images, text, or the like to a user. As another, non-limiting, example, the output device **1020** may include a wearable display separate from, but communicatively coupled to, the rest of the user equipment apparatus **1000**, such as a smart watch, smart glasses, a heads-up display, or the like. Further, the output device **1020** may be a component of a smart phone, a personal digital assistant, a television, a table computer, a notebook (laptop) computer, a personal computer, a vehicle dashboard, or the like.

(195) In certain embodiments, the output device **1020** includes one or more speakers for producing

sound. For example, the output device **1020** may produce an audible alert or notification (e.g., a beep or chime). In some embodiments, the output device **1020** includes one or more haptic devices for producing vibrations, motion, or other haptic feedback. In some embodiments, all, or portions of the output device **1020** may be integrated with the input device **1015**. For example, the input device **1015** and output device **1020** may form a touchscreen or similar touch-sensitive display. In other embodiments, the output device **1020** may be located near the input device **1015**.

(196) The transceiver **1025** communicates with one or more network functions of a mobile communication network via one or more access networks. The transceiver **1025** operates under the control of the processor **1005** to transmit messages, data, and other signals and also to receive messages, data, and other signals. For example, the processor **1005** may selectively activate the transceiver **1025** (or portions thereof) at particular times in order to send and receive messages.

(197) The transceiver **1025** includes at least transmitter **1030** and at least one receiver **1035**. One or more transmitters **1030** may be used to provide UL communication signals to a base unit **121**, such as the UL transmissions described herein. Similarly, one or more receivers **1035** may be used to receive DL communication signals from the base unit **121**, as described herein.

(198) Although only one transmitter **1030** and one receiver **1035** are illustrated, the user equipment apparatus **1000** may have any suitable number of transmitters **1030** and receivers **1035**. Further, the transmitter(s) **1030** and the receiver(s) **1035** may be any suitable type of transmitters and receivers. In one embodiment, the transceiver **1025** includes a first transmitter/receiver pair used to communicate with a mobile communication network over licensed radio spectrum and a second transmitter/receiver pair used to communicate with a mobile communication network over unlicensed radio spectrum.

(199) In certain embodiments, the first transmitter/receiver pair used to communicate with a mobile communication network over licensed radio spectrum and the second transmitter/receiver pair used to communicate with a mobile communication network over unlicensed radio spectrum may be combined into a single transceiver unit, for example a single chip performing functions for use with both licensed and unlicensed radio spectrum. In some embodiments, the first transmitter/receiver pair and the second transmitter/receiver pair may share one or more hardware components. For example, certain transceivers **1025**, transmitters **1030**, and receivers **1035** may be implemented as physically separate components that access a shared hardware resource and/or software resource, such as for example, the network interface **1040**.

(200) In various embodiments, one or more transmitters **1030** and/or one or more receivers **1035** may be implemented and/or integrated into a single hardware component, such as a multi-transceiver chip, a system-on-a-chip, an Application-Specific Integrated Circuit ("ASIC"), or other type of hardware component. In certain embodiments, one or more transmitters **1030** and/or one or more receivers **1035** may be implemented and/or integrated into a multi-chip module. In some embodiments, other components such as the network interface **1040** or other hardware components/circuits may be integrated with any number of transmitters **1030** and/or receivers **1035** into a single chip. In such embodiment, the transmitters **1030** and receivers **1035** may be logically configured as a transceiver **1025** that uses one more common control signals or as modular transmitters **1030** and receivers **1035** implemented in the same hardware chip or in a multi-chip module.

(201) FIG. **11** depicts a network apparatus **1100** that may be used for configuring positioning measurements and reports, according to embodiments of the disclosure. In one embodiment, network apparatus **1100** may be one implementation of a RAN node, such as the base unit **121** and/or the RAN node **210**, as described above. Furthermore, the base network apparatus **1100** may include a processor **1105**, a memory **1110**, an input device **1115**, an output device **1120**, and a transceiver **1125**.

(202) In some embodiments, the input device **1115** and the output device **1120** are combined into a single device, such as a touchscreen. In certain embodiments, the network apparatus **1100** may not

include any input device **1115** and/or output device **1120**. In various embodiments, the network apparatus **1100** may include one or more of: the processor **1105**, the memory **1110**, and the transceiver **1125**, and may not include the input device **1115** and/or the output device **1120**.
(203) As depicted, the transceiver **1125** includes at least one transmitter **1130** and at least one receiver **1135**. Here, the transceiver **1125** communicates with one or more remote units **175**. Additionally, the transceiver **1125** may support at least one network interface **1140** and/or application interface **1145**. The application interface(s) **1145** may support one or more APIs. The network interface(s) **1140** may support 3GPP reference points, such as Uu, N1, N2 and N3. Other network interfaces **1140** may be supported, as understood by one of ordinary skill in the art.
(204) The processor **1105**, in one embodiment, may include any known controller capable of executing computer-readable instructions and/or capable of performing logical operations. For example, the processor **1105** may be a microcontroller, a microprocessor, a CPU, a GPU, an auxiliary processing unit, a FPGA, or similar programmable controller. In some embodiments, the processor **1105** executes instructions stored in the memory **1110** to perform the methods and routines described herein. The processor **1105** is communicatively coupled to the memory **1110**, the input device **1115**, the output device **1120**, and the transceiver **1125**.
(205) In various embodiments, the network apparatus **1100** is a RAN node (e.g., gNB) that communicates with one or more UEs, as described herein. In such embodiments, the processor **1105** controls the network apparatus **1100** to perform the above described RAN behaviors. When operating as a RAN node, the processor **1105** may include an application processor (also known as “main processor”) which manages application-domain and operating system (“OS”) functions and a baseband processor (also known as “baseband radio processor”) which manages radio functions.
(206) In various embodiments, the processor **1105** and transceiver **1125** control the network apparatus **1100** to perform the above described LMF behaviors. For example, in one embodiment, the transceiver **1125** sends, to a User Equipment (“UE”) device, a positioning configuration defining a positioning configuration timeline and a measurement and processing time window for the UE. In one embodiment, the positioning configuration comprising a timeline duration defining when to start performing measurements, a set of positioning measurements to be taken within the configured time window, and a window duration for measuring and processing requested position-relation measurements for the UE according to the positioning processing timeline.
(207) In one embodiment, the transceiver **11215** receives, from the UE device, a positioning measurement report comprising the at least one positioning measurement and measurement timeline of the at least one positioning measurement performed within the configured time window. In one embodiment, the positioning configuration timeline is determined as a function of different individual timelines associated with configuration of assistance data, measurements, processing, and calculation of the UE's position estimate.
(208) In one embodiment, the transceiver **1125** sends, to a User Equipment (“UE”) device, an uplink (“UL”) configured grant configuration based on criteria associated with at least one of a measurement priority order, a positioning latency budget, and a positioning processing timeline for the UE and receives a positioning measurement report from the UE using the UL configured grant configuration based on an availability of positioning-related reference signal measurements based on at least one of the measurement priority order and the positioning processing timeline.
(209) The memory **1110**, in one embodiment, is a computer readable storage medium. In some embodiments, the memory **1110** includes volatile computer storage media. For example, the memory **1110** may include a RAM, including dynamic RAM (“DRAM”), synchronous dynamic RAM (“SDRAM”), and/or static RAM (“SRAM”). In some embodiments, the memory **1110** includes non-volatile computer storage media. For example, the memory **1110** may include a hard disk drive, a flash memory, or any other suitable non-volatile computer storage device. In some embodiments, the memory **1110** includes both volatile and non-volatile computer storage media.
(210) In some embodiments, the memory **1110** stores data related to configuring positioning

measurements and reports. For example, the memory **1110** may store parameters, configurations, resource assignments, policies, and the like, as described above. In certain embodiments, the memory **1110** also stores program code and related data, such as an operating system or other controller algorithms operating on the apparatus **1100**.

(211) The input device **1115**, in one embodiment, may include any known computer input device including a touch panel, a button, a keyboard, a stylus, a microphone, or the like. In some embodiments, the input device **1115** may be integrated with the output device **1120**, for example, as a touchscreen or similar touch-sensitive display. In some embodiments, the input device **1115** includes a touchscreen such that text may be input using a virtual keyboard displayed on the touchscreen and/or by handwriting on the touchscreen. In some embodiments, the input device **1115** includes two or more different devices, such as a keyboard and a touch panel.

(212) The output device **1120**, in one embodiment, is designed to output visual, audible, and/or haptic signals. In some embodiments, the output device **1120** includes an electronically controllable display or display device capable of outputting visual data to a user. For example, the output device **1120** may include, but is not limited to, an LCD display, an LED display, an OLED display, a projector, or similar display device capable of outputting images, text, or the like to a user. As another, non-limiting, example, the output device **1120** may include a wearable display separate from, but communicatively coupled to, the rest of the network apparatus **1100**, such as a smart watch, smart glasses, a heads-up display, or the like. Further, the output device **1120** may be a component of a smart phone, a personal digital assistant, a television, a table computer, a notebook (laptop) computer, a personal computer, a vehicle dashboard, or the like.

(213) In certain embodiments, the output device **1120** includes one or more speakers for producing sound. For example, the output device **1120** may produce an audible alert or notification (e.g., a beep or chime). In some embodiments, the output device **1120** includes one or more haptic devices for producing vibrations, motion, or other haptic feedback. In some embodiments, all, or portions of the output device **1120** may be integrated with the input device **1115**. For example, the input device **1115** and output device **1120** may form a touchscreen or similar touch-sensitive display. In other embodiments, the output device **1120** may be located near the input device **1115**.

(214) The transceiver **1125** includes at least transmitter **1130** and at least one receiver **1135**. One or more transmitters **1130** may be used to communicate with the UE, as described herein. Similarly, one or more receivers **1135** may be used to communicate with network functions in the PLMN and/or RAN, as described herein. Although only one transmitter **1130** and one receiver **1135** are illustrated, the network apparatus **1100** may have any suitable number of transmitters **1130** and receivers **1135**. Further, the transmitter(s) **1130** and the receiver(s) **1135** may be any suitable type of transmitters and receivers.

(215) FIG. **12** depicts one embodiment of a method **1200** for configuring positioning measurements and reports, according to embodiments of the disclosure. In various embodiments, the method **1200** is performed by a user equipment device in a mobile communication network, such as the remote unit **105**, the UE **205**, and/or the user equipment apparatus **1000**, described above. In some embodiments, the method **1200** is performed by a processor, such as a microcontroller, a microprocessor, a CPU, a GPU, an auxiliary processing unit, a FPGA, or the like.

(216) In one embodiment, the method **1200** begins and receives **1205**, from a mobile wireless communication network, a positioning configuration defining a positioning configuration timeline and a measurement and processing time window for the UE. The method **1200**, in some embodiments, includes performing **1210** at least one positioning measurement for the UE according to the positioning processing timeline in response to receiving the positioning configuration. The method **1200**, in further embodiments, includes sending **1215** a positioning measurement report comprising the at least one positioning measurement and measurement timeline performed of the at least one positioning measurement within the configured time window from the UE to the mobile wireless communication network, and the method **1200** ends.

(217) FIG. **13** depicts one embodiment of a method **1300** for configuring positioning measurements and reports, according to embodiments of the disclosure. In various embodiments, the method **1300** is performed by a Location Management Function in a mobile communication network, such as the LMF **144**, and/or the network apparatus **1100**, described above. In some embodiments, the method **1300** is performed by a processor, such as a microcontroller, a microprocessor, a CPU, a GPU, an auxiliary processing unit, a FPGA, or the like.
(218) The method **1300** begins and sends **1305**, in one embodiment, to a UE, a positioning configuration defining a positioning configuration timeline and a measurement and processing time window for the UE. The method **1300**, in one embodiment, includes receiving **1310**, from the UE device, a positioning measurement report comprising the at least one positioning measurement and measurement timeline of the at least one positioning measurement performed within the configured time window, and the method **1300** ends.
(219) FIG. **14** depicts one embodiment of a method **1400** for configuring positioning measurements and reports, according to embodiments of the disclosure. In various embodiments, the method **1400** is performed by a user equipment device in a mobile communication network, such as the remote unit **105**, the UE **205**, and/or the user equipment apparatus **1000**, described above. In some embodiments, the method **1400** is performed by a processor, such as a microcontroller, a microprocessor, a CPU, a GPU, an auxiliary processing unit, a FPGA, or the like.
(220) In one embodiment, the method **1400** begins and receives **1405**, from a mobile wireless communication network, an uplink (“UL”) configured grant configuration based on criteria associated with at least one of a measurement priority order, a positioning latency budget, and a positioning processing timeline for the UE. In one embodiment, the method **1400** performs **1410** at least one positioning measurement for the UE according to at least one of the measurement priority order and the positioning processing timeline and generates **1415** a positioning measurement report comprising the at least one positioning measurement.
(221) In certain embodiments, the method **1400** sends **1420** the positioning measurement report to the mobile wireless communication network using the UL configured grant configuration based on an availability of positioning-related reference signal measurements based on at least one of the measurement priority order and the positioning processing timeline, and the method **1400** ends.
(222) FIG. **15** depicts one embodiment of a method **1300** for configuring positioning measurements and reports, according to embodiments of the disclosure. In various embodiments, the method **1300** is performed by a Location Management Function in a mobile communication network, such as the LMF **144**, and/or the network apparatus **1100**, described above. In some embodiments, the method **1300** is performed by a processor, such as a microcontroller, a microprocessor, a CPU, a GPU, an auxiliary processing unit, a FPGA, or the like.
(223) In one embodiment, the method **1500** begins and sends **1505**, to a User Equipment (“UE”) device, an uplink (“UL”) configured grant configuration based on criteria associated with at least one of a measurement priority order, a positioning latency budget, and a positioning processing timeline for the UE. In further embodiments, the method **1500** receives **1510** a positioning measurement report from the UE device using the UL configured grant configuration based on an availability of positioning-related reference signal measurements based on at least one of the measurement priority order and the positioning processing timeline, and the method **1500** ends.
(224) Disclosed herein is a first apparatus for configuring positioning measurements and reports, according to embodiments of the disclosure. The first apparatus may be implemented by a user equipment device in a mobile communication network, such as the remote unit **105**, the UE **205**, and/or the user equipment apparatus **1000**, described above.
(225) In one embodiment, the first apparatus includes a transceiver that receives, from a mobile wireless communication network, a positioning configuration defining a positioning configuration timeline and a measurement and processing time window for the UE. The positioning configuration may include a timeline duration defining when to start performing measurements, a set of

positioning measurements to be taken within the configured time window, and a window duration for measuring and processing requested position-relation measurements for the UE according to the positioning processing timeline.
(226) In one embodiment, the first apparatus includes a processor that performs at least one positioning measurement for the UE according to the positioning processing timeline in response to receiving the positioning configuration. In certain embodiments, the transceiver sends a positioning measurement report comprising the at least one positioning measurement and measurement timeline performed of the at least one positioning measurement within the configured time window from the UE to the mobile wireless communication network.
(227) In one embodiment, multiple one of configuration timelines and measurement and processing timelines may be configured for the UE, and wherein a latency for at least one of a reference signal received power (“RSRP”), a reference signal time difference (“RSTD”), and a UE Rx-Tx positioning measurement within each of the multiple measurement and processing timelines may be reported to the mobile wireless communication network.
(228) In one embodiment, the latency may comprise a single value in response to positioning measurements being performed at a same time and multiple values in response to positioning measurements being performed sequentially.
(229) In one embodiment, the transceiver receives pre-configured assistance data for positioning from the mobile wireless communication network during a Long-term evolution Protocol Positioning (“LPP”) session and performs measurements and processing on the pre-configured assistance data in response to the transceiver receiving an LPP Request Location Information message.
(230) In one embodiment, the transceiver receives the positioning configuration from the mobile wireless communication network via a broadcast signal, the positioning configuration designed for a plurality of UEs with same capabilities.
(231) In one embodiment, the transceiver receives the positioning configuration from the mobile wireless communication network via a UE-specific dedicated signal, the UE-specific dedicated signal comprising an LPP Request Location Information message.
(232) In one embodiment, in response to the UE initiating a positioning reference signal (“PRS”) configuration request, the processor determines an overall timeline between receiving the configuration request and receiving the UE's position estimate, the overall timeline comprising multiple timelines related to the configuration of assistance data, measurement, processing, and calculation of the UE's position estimate.
(233) In one embodiment, the transceiver sends an on-demand request to the mobile wireless communication network to receive downlink-PRS (“DL-PRS”) assistance data in response to at least one of no prior DL-PRS physical layer configuration stored at the UE, existing DL-PRS configuration is outdated, and the existing DL-PRS configuration does not meet accuracy requirements.
(234) In one embodiment, the positioning configuration further comprises a set of measurement gap configurations to be applied by the UE for the positioning processing timeline, the measurement gap configuration defining a measurement gap length and a measurement gap length and a measurement gap repetition period for the positioning processing timeline.
(235) In one embodiment, the set of measurement gap configurations can be pre-configured in the UE via signaling from the mobile wireless communication network. In one embodiment, the processor processes PRS measurements and reports according to a UE PRS processing unit (“PPU”), the UE PPU comprising a number of PPUs that the UE can support for a given symbol.
(236) In one embodiment, based on the UE's capabilities, the processor performs parallel processing of the same PRS symbol with respect to other positioning measurements. In one embodiment, the mobile wireless communication network comprises at least one of a base station and a location management function.

(237) Disclosed herein is a first method for configuring positioning measurements and reports, according to embodiments of the disclosure. The first method may be performed by a user equipment device in a mobile communication network, such as the remote unit **105**, the UE **205**, and/or the user equipment apparatus **1000**, described above.
(238) In one embodiment, the first method includes receiving, from a mobile wireless communication network, a positioning configuration defining a positioning configuration timeline and a measurement and processing time window for the UE. The positioning configuration may include a timeline duration defining when to start performing measurements, a set of positioning measurements to be taken within the configured time window, and a window duration for measuring and processing requested position-relation measurements for the UE according to the positioning processing timeline.
(239) In one embodiment, the first method includes performing at least one positioning measurement for the UE according to the positioning processing timeline in response to receiving the positioning configuration. In certain embodiments, the first method includes sending a positioning measurement report comprising the at least one positioning measurement and measurement timeline performed of the at least one positioning measurement within the configured time window from the UE to the mobile wireless communication network.
(240) In one embodiment, multiple one of configuration timelines and measurement and processing timelines may be configured for the UE, and wherein a latency for at least one of a reference signal received power (“RSRP”), a reference signal time difference (“RSTD”), and a UE Rx-Tx positioning measurement within each of the multiple measurement and processing timelines may be reported to the mobile wireless communication network.
(241) In one embodiment, the latency may comprise a single value in response to positioning measurements being performed at a same time and multiple values in response to positioning measurements being performed sequentially.
(242) In one embodiment, the first method includes receiving pre-configured assistance data for positioning from the mobile wireless communication network during a Long-term evolution Protocol Positioning (“LPP”) session and performs measurements and processing on the pre-configured assistance data in response to the transceiver receiving an LPP Request Location Information message.
(243) In one embodiment, the first method includes receiving the positioning configuration from the mobile wireless communication network via a broadcast signal, the positioning configuration designed for a plurality of UEs with same capabilities.
(244) In one embodiment, the first method includes receiving the positioning configuration from the mobile wireless communication network via a UE-specific dedicated signal, the UE-specific dedicated signal comprising an LPP Request Location Information message.
(245) In one embodiment, in response to the UE initiating a positioning reference signal (“PRS”) configuration request, the method includes determining an overall timeline between receiving the configuration request and receiving the UE's position estimate, the overall timeline comprising multiple timelines related to the configuration of assistance data, measurement, processing, and calculation of the UE's position estimate.
(246) In one embodiment, the first method includes sending an on-demand request to the mobile wireless communication network to receive downlink-PRS (“DL-PRS”) assistance data in response to at least one of no prior DL-PRS physical layer configuration stored at the UE, existing DL-PRS configuration is outdated, and the existing DL-PRS configuration does not meet accuracy requirements.
(247) In one embodiment, the positioning configuration further comprises a set of measurement gap configurations to be applied by the UE for the positioning processing timeline, the measurement gap configuration defining a measurement gap length and a measurement gap length and a measurement gap repetition period for the positioning processing timeline.

(248) In one embodiment, the set of measurement gap configurations can be pre-configured in the UE via signaling from the mobile wireless communication network. In one embodiment, the first method includes processing PRS measurements and reports according to a UE PRS processing unit ("PPU"), the UE PPU comprising a number of PPUs that the UE can support for a given symbol.
(249) In one embodiment, based on the UE's capabilities, the first method includes performing parallel processing of the same PRS symbol with respect to other positioning measurements. In one embodiment, the mobile wireless communication network comprises at least one of a base station and a location management function.
(250) Disclosed herein is a second apparatus for configuring positioning measurements and reports, according to embodiments of the disclosure. The second apparatus may be implemented by a base station, e.g., a gNB, a location management function in a mobile communication network, such as the LMF **144**, and/or the network apparatus **1100**, described above.
(251) In one embodiment, the second apparatus includes a transceiver that sends, to a User Equipment ("UE") device, a positioning configuration defining a positioning configuration timeline and a measurement and processing time window for the UE. In one embodiment, the positioning configuration comprising a timeline duration defining when to start performing measurements, a set of positioning measurements to be taken within the configured time window, and a window duration for measuring and processing requested position-relation measurements for the UE according to the positioning processing timeline.
(252) In one embodiment, the transceiver receives, from the UE device, a positioning measurement report comprising the at least one positioning measurement and measurement timeline of the at least one positioning measurement performed within the configured time window. In one embodiment, the positioning configuration timeline is determined as a function of different individual timelines associated with configuration of assistance data, measurements, processing, and calculation of the UE's position estimate.
(253) Disclosed herein is a second method for configuring positioning measurements and reports, according to embodiments of the disclosure. The second method may be performed by a base station, e.g., a gNB, a location management function device in a mobile communication network, such as the LMF **144**, and/or the network apparatus **1700**, described above.
(254) In one embodiment, the second method includes sending, to a User Equipment ("UE") device, a positioning configuration defining a positioning configuration timeline and a measurement and processing time window for the UE. In one embodiment, the positioning configuration comprising a timeline duration defining when to start performing measurements, a set of positioning measurements to be taken within the configured time window, and a window duration for measuring and processing requested position-relation measurements for the UE according to the positioning processing timeline.
(255) In one embodiment, the second method includes receiving, from the UE device, a positioning measurement report comprising the at least one positioning measurement and measurement timeline of the at least one positioning measurement performed within the configured time window. In one embodiment, the positioning configuration timeline is determined as a function of different individual timelines associated with configuration of assistance data, measurements, processing, and calculation of the UE's position estimate.
(256) Disclosed herein is a third apparatus for configuring positioning measurements and reports, according to embodiments of the disclosure. The third apparatus may be implemented by a user equipment device in a mobile communication network, such as the remote unit **105**, the UE **205**, and/or the user equipment apparatus **1600**, described above.
(257) In one embodiment, the third apparatus includes a transceiver that receives, from a mobile wireless communication network, an uplink ("UL") configured grant configuration based on criteria associated with at least one of a measurement priority order, a positioning latency budget, and a positioning processing timeline for the UE. In some embodiments, the third apparatus

includes a processor that performs at least one positioning measurement for the UE according to at least one of the measurement priority order and the positioning processing timeline and generates a positioning measurement report comprising the at least one positioning measurement.

(258) In certain embodiments, the transceiver sends the positioning measurement report to the mobile wireless communication network using the UL configured grant configuration based on an availability of positioning-related reference signal measurements based on at least one of the measurement priority order and the positioning processing timeline.

(259) In one embodiment, the UE is configured with a Type 1 UL configured grant configuration via radio resource control ("RRC") signaling for sending the positioning measurement report comprising the at least one positioning measurement.

(260) In one embodiment, the UE is configured with a Type 2 UL configured grant configuration via downlink control information ("DCI") signaling for sending the positioning measurement report comprising the at least one positioning measurement.

(261) In one embodiment, the UL configured grant configuration comprises signaling information for one or more of an offset, a periodicity, an activation, a deactivation, and time-frequency resources of the positioning measurement report comprising the at least one positioning measurement.

(262) In one embodiment, the transceiver receives a positioning measurement configuration indicating the positioning measurement priority order based on the availability of positioning-related reference signal resources, the priority order determined based on at least one of the UE's capabilities, the positioning latency budget, and a location estimate accuracy. In one embodiment, the processor performs the at least one positioning measurement and generates the positioning measurement report according to the priority order indicated in the positioning measurement configuration. In one embodiment, the transceiver sends the positioning measurement report to the mobile wireless communication network using the UL configured grant configuration according to the priority order indicated in the measurement configuration.

(263) In one embodiment, the transceiver receives the positioning measurement configuration in response to sending a request for the positioning measurement configuration. In one embodiment, the positioning measurement priority order is configured by at least one of a location server and a base station of the mobile wireless communication network. In one embodiment, the location server exchanges information with the base station related to the configuration and scheduling of the physical uplink shared channel ("PUSCH").

(264) In one embodiment, the transceiver dynamically provides the positioning measurement configuration, on-demand, using at least one of radio resource control ("RRC") signaling and medium access control ("MAC") control element ("CE") signaling. In one embodiment, the processor drops the positioning-related reference signal measurements to be reported in response to at least one of a measurement report size exceeding UL transmission resource availability, measurements being lower in priority with respect to other measurements with higher priority, measurements being one of incomplete and corrupted, and measurements determined to be unreliable in response to not satisfying one or more integrity requirements.

(265) In one embodiment, the processor drops the positioning-related reference signal measurements based on at least one of the latency budget and the measurement and processing timeline.

(266) Disclosed herein is a third method for configuring positioning measurements and reports, according to embodiments of the disclosure. The third method may be performed by a user equipment device in a mobile communication network, such as the remote unit **105**, the UE **205**, and/or the user equipment apparatus **1600**, described above.

(267) In one embodiment, the third method includes receiving, from a mobile wireless communication network, an uplink ("UL") configured grant configuration based on criteria associated with at least one of a measurement priority order, a positioning latency budget, and a

positioning processing timeline for the UE. In some embodiments, the third method includes performing at least one positioning measurement for the UE according to at least one of the measurement priority order and the positioning processing timeline and generating a positioning measurement report comprising the at least one positioning measurement.

(268) In certain embodiments, the third method includes sending the positioning measurement report to the mobile wireless communication network using the UL configured grant configuration based on an availability of positioning-related reference signal measurements based on at least one of the measurement priority order and the positioning processing timeline.

(269) In one embodiment, the third method includes configuring the UE with a Type 1 UL configured grant configuration via radio resource control (“RRC”) signaling for sending the positioning measurement report comprising the at least one positioning measurement.

(270) In one embodiment, the third method includes configuring the UE with a Type 2 UL configured grant configuration via downlink control information (“DCI”) signaling for sending the positioning measurement report comprising the at least one positioning measurement.

(271) In one embodiment, the UL configured grant configuration comprises signaling information for one or more of an offset, a periodicity, an activation, a deactivation, and time-frequency resources of the positioning measurement report comprising the at least one positioning measurement.

(272) In one embodiment, the third method includes receiving a positioning measurement configuration indicating a positioning measurement priority order based on the availability of positioning-related reference signal resources, the priority order determined based on at least one of the UE's capabilities, the positioning latency budget, and a location estimate accuracy. In one embodiment, the third method includes performing the at least one positioning measurement and generates the positioning measurement report according to the priority order indicated in the positioning measurement configuration. In one embodiment, the third method includes sending the positioning measurement report to the mobile wireless communication network using the UL configured grant configuration according to the priority order indicated in the measurement configuration.

(273) In one embodiment, the third method includes receiving the positioning measurement configuration in response to sending a request for the positioning measurement configuration. In one embodiment, the positioning measurement priority order is configured by at least one of a location server and a base station of the mobile wireless communication network. In one embodiment, the location server exchanges information with the base station related to the configuration and scheduling of the physical uplink shared channel (“PUSCH”).

(274) In one embodiment, the third method includes dynamically providing the positioning measurement configuration, on-demand, using at least one of radio resource control (“RRC”) signaling and medium access control (“MAC”) control element (“CE”) signaling. In one embodiment, the third method includes dropping the positioning-related reference signal measurements to be reported in response to at least one of a measurement report size exceeding UL transmission resource availability, measurements being lower in priority with respect to other measurements with higher priority, measurements being one of incomplete and corrupted, and measurements determined to be unreliable in response to not satisfying one or more integrity requirements.

(275) In one embodiment, the third method includes dropping the positioning-related reference signal measurements based on at least one of the latency budget and the measurement and processing timeline.

(276) Disclosed herein is a fourth apparatus for configuring positioning measurements and reports, according to embodiments of the disclosure. The fourth apparatus may be implemented by a location management function in a mobile communication network, such as the LMF **144** and/or the network apparatus **1100**, described above.

(277) In one embodiment, the fourth apparatus includes a transceiver that sends, to a User Equipment ("UE") device, an uplink ("UL") configured grant configuration based on criteria associated with at least one of a measurement priority order, a positioning latency budget, and a positioning processing timeline for the UE and receives a positioning measurement report from the UE device using the UL configured grant configuration based on an availability of positioning-related reference signal measurements based on at least one of the measurement priority order and the positioning processing timeline.
(278) Disclosed herein is a fourth method for configuring positioning measurements and reports, according to embodiments of the disclosure. The fourth method may be performed by a location management function device in a mobile communication network, such as the LMF **144** and/or the network apparatus **1100**, described above.
(279) In one embodiment, the fourth method includes sending, to a User Equipment ("UE") device, an uplink ("UL") configured grant configuration based on criteria associated with at least one of a measurement priority order, a positioning latency budget, and a positioning processing timeline for the UE and receiving a positioning measurement report from the UE device using the UL configured grant configuration based on an availability of positioning-related reference signal measurements based on at least one of the measurement priority order and the positioning processing timeline.
(280) Embodiments may be practiced in other specific forms. The described embodiments are to be considered in all respects only as illustrative and not restrictive. The scope of the invention is, therefore, indicated by the appended claims rather than by the foregoing description. All changes which come within the meaning and range of equivalency of the claims are to be embraced within their scope.

Claims

1. A apparatus, comprising: a processor; and a memory coupled with the processor, the memory comprising instructions executable by the processor to cause the apparatus to: receive, from a mobile wireless communication network, a positioning configuration defining a positioning configuration timeline and a measurement and processing time window for a user equipment ("UE"), the positioning configuration comprising a timeline duration defining when to start performing measurements, a set of positioning measurements to be taken within the configured time window, and a window duration for measuring and processing requested position-relation measurements for the UE according to the positioning processing timeline; perform at least one positioning measurement for the UE according to the positioning processing timeline in response to receiving the positioning configuration; and send a positioning measurement report comprising the at least one positioning measurement and measurement timeline performed of the at least one positioning measurement within the configured time window from the UE to the mobile wireless communication network.
2. The apparatus of claim 1, wherein multiple one of configuration timelines and measurement and processing timelines may be configured for the UE, and wherein at least one of a reference signal received power ("RSRP"), a reference signal time difference ("RSTD"), and a UE Rx-Tx positioning measurement within each of the multiple measurement and processing timelines may be performed and reported to the mobile wireless communication network.
3. The apparatus of claim 2, wherein a positioning reference signal ("PRS") measurement and processing timeline duration for the UE may comprise: a single value in response to positioning measurements being performed at a same time; and multiple values in response to positioning measurements being performed sequentially.
4. The apparatus of claim 1, wherein the instructions are further executable by the processor to cause the apparatus to receive pre-configured assistance data for positioning from the mobile

wireless communication network during a Long-term evolution Protocol Positioning (“LPP”) session and perform measurements and processing on the pre-configured assistance data in response to the transceiver receiving an LPP Request Location Information message.
5. The apparatus of claim 1, wherein the instructions are further executable by the processor to cause the apparatus to receive the positioning configuration from the mobile wireless communication network via a broadcast signal, the positioning configuration designed for a plurality of UEs with same capabilities.
6. The apparatus of claim 1, wherein the instructions are further executable by the processor to cause the apparatus to receive the positioning configuration from the mobile wireless communication network via a UE-specific dedicated signal, the UE-specific dedicated signal comprising a medium access control (“MAC”) control element (“CE”) or radio resource control (“RRC”) signaling message.
7. The apparatus of claim 1, wherein the instructions are further executable by the processor to cause the apparatus to the transceiver send an on-demand request to the mobile wireless communication network to receive downlink positioning reference signal (“DL-PRS”) assistance data in response to at least one of: no prior DL-PRS physical layer configuration stored at the UE; existing DL-PRS configuration is outdated; and the existing DL-PRS configuration does not meet accuracy requirements.
8. The apparatus of claim 1, wherein the positioning configuration further comprises a set of measurement gap configurations to be applied by the UE for the positioning processing timeline, the measurement gap configuration defining a measurement gap length and a measurement gap length and a measurement gap repetition period for the positioning processing timeline.
9. The apparatus of claim 8, wherein the set of measurement gap configurations can be pre-configured in the UE via signaling from the mobile wireless communication network.
10. The apparatus of claim 1, wherein the instructions are further executable by the processor to cause the apparatus to process positioning-related measurements based on different positioning timeline configurations associated with different UE capabilities.
11. The apparatus of claim 10, wherein, based on the UE's capabilities, the instructions are further executable by the processor to cause the apparatus to process N downlink (“DL”) positioning reference signal (“PRS”) symbols every T ms for a given maximum bandwidth supported by the UE according to the positioning timeline configuration within an active DL bandwidth part (“BWP”) configuration.
12. The apparatus of claim 1, wherein the mobile wireless communication network comprises a base station or a location management function.
13. A method, comprising: receiving, from a mobile wireless communication network, a positioning configuration defining a positioning configuration timeline and a, measurement and processing time window for a user equipment (“UE”), the positioning configuration comprising a timeline duration defining when to start performing measurements, a set of positioning measurements to be made taken within the configured time window, and a window duration for measuring and processing requested position-relation measurements for the UE according to the positioning processing timeline; performing at least one positioning measurement for the UE according to the positioning processing timeline in response to receiving the positioning configuration; and sending a positioning measurement report comprising the at least one positioning measurement and measurement timeline performed of the at least one positioning measurement within the configured time window from the UE to the mobile wireless communication network.
14. The method of claim 13, wherein multiple one of configuration timelines and measurement and processing timelines may be configured for the UE, and wherein at least one of a reference signal received power (“RSRP”), a reference signal time difference (“RSTD”), and a UE Rx-Tx positioning measurement within each of the multiple measurement and processing timelines may

be performed and reported to the mobile wireless communication network.
15. The method of claim 14, wherein a positioning reference signal (“PRS”) measurement and processing timeline duration for the UE may comprise: a single value in response to positioning measurements being performed at a same time; and multiple values in response to positioning measurements being performed sequentially.
16. The method of claim 13, further comprising receiving pre-configured assistance data for positioning from the mobile wireless communication network during a Long- term evolution Protocol Positioning (“LPP”) session and performing measurements and processing on the pre-configured assistance data in response to the transceiver receiving an LPP Request Location Information message.
17. The method of claim 13, further comprising receiving the positioning configuration from the mobile wireless communication network via a broadcast signal, the positioning configuration designed for a plurality of UEs with same capabilities.
18. The method of claim 13, further comprising receiving the positioning configuration from the mobile wireless communication network via a UE-specific dedicated signal, the UE-specific dedicated signal comprising a medium access control (“MAC”) control element (“CE”) or radio resource control (“RRC”) signaling message.
19. An apparatus, comprising: a processor; and a memory coupled with the processor, the memory comprising instructions executable by the processor to cause the apparatus to: send, to a User Equipment (“UE”) device, a positioning configuration defining a positioning configuration timeline and a measurement and processing time window for the UE, the positioning configuration comprising a timeline duration defining when to start performing measurements, a set of positioning measurements to be taken within the configured time window, and a window duration for measuring and processing requested position-relation measurements for the UE according to the positioning processing timeline; and receive, from the UE device, a positioning measurement report comprising the at least one positioning measurement and measurement timeline of the at least one positioning measurement performed within the configured time window.
20. The apparatus of claim 19, wherein the positioning configuration timeline is determined as a function of different individual timelines associated with configuration of assistance data, measurements, processing, and calculation of the UE's position estimate.